Current-Limiting Control Design for Grid-Forming Capability Enhancement of IBRs Under Asymmetric Grid Disturbances

Qian Liu, Miao Wang, Mostafa Nick, Cheng Chen, and Xiaowei Zhao, *Member, IEEE*

***Abstract*—Recent technical specifications for grid-forming (GFM) inverter-based resources (IBRs) require the terminal voltage to remain as balanced as possible during asymmetric disturbances to enhance GFM capability. However, under severe unbalanced fault conditions, GFM IBRs are forced into current-limiting operation, which substantially impairs their ability to support voltage balance. This presents a critical challenge for advanced current-limiting control (CLC) in GFM IBRs. To address this challenge, a general CLC design guideline is proposed. The core principle is to align the phase angle of the impedance emulated by the CLC with that of the external AC grid impedance in both positive and negative sequences, thereby improving voltage balance and enhancing GFM capability. Following this guideline, two CLC implementations are developed using virtual impedance and current reference saturation frameworks. The proposed CLC schemes undergo rigorous grid code compliance evaluation, transient stability analysis, and robustness assessment. Extensive simulation studies, including comparisons with state-of-the-art CLC methods, as well as controller hardware-in-the-loop experiments, validate the improved transient stability and enhanced GFM capability achieved by the proposed method.**

***Index Terms*—Grid-forming inverters, current-limiting control, asymmetric faults, fault ride-through, transient stability.**

I. Introduction

DUE TO their synchronous generator (SG)-like voltage magnitude and frequency regulation capabilities, grid-forming (GFM) inverter-based resources (IBRs) are seen as indispensable components in IBR-dominated power systems [1]. A fundamental requirement for GFM IBRs is the ability to maintain terminal voltage under normal operating conditions, known as GFM capability [2].

Due to the low-bandwidth power control loops, GFM IBRs inherently maintain a nearly constant terminal voltage, making this requirement easy to fulfil under normal operations [3]. However, during severe asymmetric grid disturbances, current-limiting control (CLC) takes precedence to protect IBRs from excessive currents, and both positive- and negative-sequence components emerge [4]. These interventions induce rapid variations in IBR's terminal voltage, which significantly undermine GFM capability. Although maintaining an ideal terminal voltage is no longer possible under such conditions, system operators increasingly demand that GFM IBRs to sustain balanced terminal voltage as much as possible during unbalanced grid conditions [5]–[8]. This shifts the operational target for GFM capability under asymmetric disturbances, and the intent is to preserve the voltage source characteristics of GFM IBRs to some extent, thereby enabling natural power responses from IBRs to mitigate grid disturbances, similar to SGs [9]. From a control perspective, meeting this requirement necessitates embedding voltage support functionality within the CLC. Specifically, the CLC should maximise the positive-sequence voltage while minimise the negative-sequence voltage at the IBR terminal. By satisfying this criterion, GFM IBRs can improve the terminal voltage balance, thus, better align their performance with GFM capability expectations under unbalanced grid conditions.

Current reference saturation (CRS)-based and virtual impedance (VI)-based methods represent the two predominant CLC schemes for GFM IBRs [10]. The CRS approach directly manipulates the current setpoint during transients, whilst the VI method curtails current indirectly by actuating over the voltage setpoint. Both schemes can be implemented within either sequence-coupled or sequence-decomposed control frameworks. Under the sequence-coupled framework, the CRS-based scheme can support the terminal voltage during large disturbances by prioritising reactive current injection [11]. Similarly, the predefined X/R ratio in VI-based schemes can be tuned to enhance reactive current contribution during faults [12]. However, applying sequence-coupled control to both voltage and current controllers restricts the independent regulation of negative-sequence components. Consequently, the methods in [11] and [12] are primarily suited for symmetric grid disturbances and experience considerable performance degradation under asymmetric conditions. Typical issues include current distortion, underutilisation of the IBR's current headroom, and unconfined negative-sequence current [13].

For more complex and common asymmetric disturbances, GFM IBRs are required to independently regulate positive- and negative-sequence components, necessitating a sequence-decomposed control framework [14], [15]. Recent studies have reported CLC designs tailored for sequence-decomposed GFM IBRs. In [16], a CRS-based CLC is devised in the natural

Manuscript received 28 February, 2025; resubmitted 18 July 2025; revised 7 October 2025; accepted 10 November 2025. This work was funded by the UK Research and Innovation (UKRI) under the UK Government's Horizon Europe Funding Guarantee grant No. EP/X032477/1, which is associated with the European Union's Horizon 2020 research and innovation programme under the Marie Sklodowska-Curie grant agreement No. 101073558. (*Corresponding author: Xiaowei Zhao.*)

Qian Liu, Miao Wang, and Xiaowei Zhao are with the Intelligent Control and Smart Energy (ICSE) Research Group, School of Engineering, University of Warwick, Coventry CV4 7AL, U.K. (e-mail: qian.liu.8@warwick.ac.uk; miao.wang. 4@warwick.ac.uk; xiaowei.zhao@warwick.ac.uk).

Mostafa Nick and Cheng Chen are with the Offline Network Modelling team, National Energy System Operator (NESO), Warwick CV34 6DA, U.K. (e-mail: mostafa.nick@neso.energy; cheng.chen@neso.energy).

reference frame to limit phase current magnitudes, but it risks exceeding current limits in three-phase three-wire systems. Alternative approaches, such as implementing elliptical current limiter in positive- and negative-sequence controllers [17], or recasting voltage and current controllers [18], [19], effectively constrain output current but fail to comply with grid codes (GCs) due to inefficient current utilisation and non-compliant negative-sequence current injection.

To meet the fault current injection requirements specified in recent GCs [20]–[22], hybrid CLC schemes based on sequence-decomposed control have emerged, including the integration of VI with CRS [23], virtual admittance (VA) with CRS [24], VI with voltage limiter [25], and CRS with voltage limiter [26]. By regulating the positive- and negative-sequence reactive currents of the GFM IBR against the corresponding thresholds defined in GCs, these hybrid methods can furnish voltage support in both sequences during faults. However, a clear CLC methodology that simultaneously maximises the positive-sequence voltage and minimises the negative-sequence voltage at the IBR terminal remains largely unexplored. As noted, system operators require the terminal voltage to remain as balanced as possible during asymmetric disturbances to support GFM capability. Therefore, further investigation for CLC is required to fully harness the voltage support potential of GFM IBRs under such conditions.

Simulation observations in [27] indicate that matching the phase angle of VI with the AC line impedance angle yields desirable voltage support performance under asymmetric faults. While pioneering, the findings are highly case-specific, rendering the CLC design empirical and restricting its adaptability to diverse operating conditions. The follow-up research in [28] extends these findings by offering analytical insights into the positive-sequence domain under symmetric grid conditions and quantifies the relationship between the positive-sequence terminal voltage and the virtual impedance angle. Nonetheless, underlying mechanisms in negative-sequence components remain insufficiently addressed. Moreover, both [27] and [28] lack rigorous justification for GC compliance.

To bridge the identified gaps, this study investigates improved CLC solutions to enhance the GFM capability of IBRs under asymmetric grid disturbances. The key contributions are summarised as follows:

1) Extend the theoretical analysis and CLC design guideline in [28] to account for more prevalent asymmetric disturbances by including negative-sequence components. Under the extended framework, GFM IBRs with VI-based CLC schemes can explicitly enhance their GFM capability under severe disturbances by optimising the control X/R ratios in both the positive- and negative-sequence domains.
2) Develop an improved CRS-based CLC scheme aligned with the extended design guideline. The proposed CRS-based method establishes a quantitative relationship between the emulated virtual impedance and the angle-priority setting. This relationship enables implicit enhancement of the GFM capability under grid disturbances by allowing the CRS-based scheme to adaptively adjust the phase angle of the current references.
3) Provide rigorous theoretical analysis of the proposed method from three critical perspectives: (i) compliance with modern GCs, particularly fault current injection in both sequences; (ii) transient stability, demonstrating that both VI- and CRS-based methods yield identical power-angle characteristics under the proposed design guideline; and (iii) robustness, considering scenarios where the external parameters required by the proposed method are subject to significant errors.

The rest of the paper proceeds as follows. Section II outlines the studied GFM control architecture. Section III details the CLC design guideline for GFM capability enhancement. Section IV introduces the proposed CLC schemes, followed by rigorous GC compliance, transient stability, and robustness analysis. Section V presents the simulation and controller hardware-in-the-loop (CHIL) test results, including comparative analyses with state-of-the-art CLC solutions in terms of both transient stability and GFM capability. Conclusions are drawn in Section VI.

II. Sequence-Decomposed GFM IBR with CLC

A. System Description

The system under study is shown in Fig. 1. A GFM IBR is connected to the point of common coupling (PCC) through an LC filter (R_f, L_f, and C_f) and a unit transformer. The remaining AC system is represented by the series combination of grid impedance $\boldsymbol{Z}_g = R_g + jX_g$ and an ideal AC voltage source. For simplicity, the transformer impedance is incorporated into $\boldsymbol{Z}_g$. The key variables include $\boldsymbol{v}_t$ (terminal voltage), $\boldsymbol{v}_{\text{pcc}}$ (PCC voltage), $\boldsymbol{v}_g$ (grid voltage), $\boldsymbol{i}_f$ (output current), $\boldsymbol{i}_g$ (grid current), and P/Q (active/reactive power).

The internal voltage source of the GFM IBR is governed by the outer-loop power control, comprising active power control (APC) and reactive power control (RPC), both implemented via droop control with coefficients m_p and m_q, respectively. The APC adjusts the phase angle reference θ based on the active power setpoint P_0 and nominal angular frequency ω_n, while the RPC regulates the voltage magnitude reference V_{ref} using the reactive power setpoint Q_0 and nominal voltage magnitude V_n. The inner voltage and current control adopt the sequence-decomposed framework [14], [15]. For each sequence, voltage control is typically realised via vector voltage control (VVC) [16], inversed current control (ICC) [29], or virtual admittance control (VAC) [24], which enables voltage reference tracking and generates current reference commands for the downstream vector current control (VCC) [25], [30]. The VCC then determines the modulation voltage reference $\boldsymbol{m}_g$ for the pulse-width modulation (PWM) process.

Without loss of generality, the sequence-decomposed voltage and current controllers operate in the dq domain (rotational reference frame), where subscripts d and q indicate the corresponding d- and q-axis components, respectively. Positive- and negative-sequence components are extracted using the notch filter-based sequence-decomposed module, where superscripts $+$ and $-$ denote the positive- and negative-sequence components, respectively. The complex voltage and current vectors of $\boldsymbol{v}_t$, $\boldsymbol{v}_g$, $\boldsymbol{i}_g$, and $\boldsymbol{i}_f$ are defined as $\boldsymbol{V}_{tdq}^{+(-)} = V_{td}^{+(-)} + jV_{tq}^{+(-)}$,

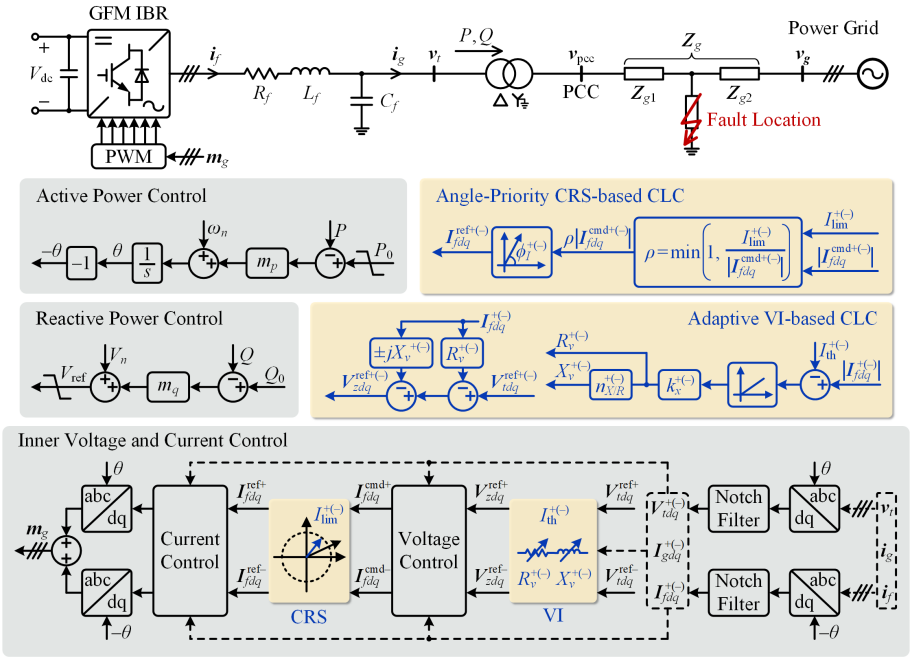


Fig. 1. Sequence-decomposed GFM IBR with current-limiting control.

$\boldsymbol{V}_{gdq}^{+(-)} = V_{gd}^{+(-)} + jV_{gq}^{+(-)}$, $\boldsymbol{I}_{gdq}^{+(-)} = I_{gd}^{+(-)} + jI_{gq}^{+(-)}$, and $\boldsymbol{I}_{fdq}^{+(-)} = I_{fd}^{+(-)} + jI_{fq}^{+(-)}$, respectively. The internal voltage references are defined as $\boldsymbol{V}_{tdq}^{\text{ref}+}$ and $\boldsymbol{V}_{tdq}^{\text{ref}-}$. The voltage controller input is $\boldsymbol{V}_{zdq}^{\text{ref}+(-)} = V_{zd}^{\text{ref}+(-)} + jV_{zq}^{\text{ref}+(-)}$, and the corresponding output current command is $\boldsymbol{I}_{fdq}^{\text{cmd}+(-)} = I_{fd}^{\text{cmd}+(-)} + jI_{fq}^{\text{cmd}+(-)}$. The input to the current controller is given by $\boldsymbol{I}_{fdq}^{\text{ref}+(-)} = I_{fd}^{\text{ref}+(-)} + jI_{fq}^{\text{ref}+(-)}$. Positive- (negative-) sequence phasors of $\boldsymbol{v}_t$, $\boldsymbol{v}_g$, $\boldsymbol{i}_g$, and $\boldsymbol{i}_f$ are denoted as $\dot{\boldsymbol{V}}_t^{+(-)}$, $\dot{\boldsymbol{V}}_g^{+(-)}$, $\dot{\boldsymbol{I}}_g^{+(-)}$, and $\dot{\boldsymbol{I}}_f^{+(-)}$, respectively.

B. Current-Limiting Control

The widely adopted VI- and CRS-based CLC schemes include the adaptive VI method, detailed in [31], and the angle-priority CRS method, detailed in [32], as illustrated in Fig. 1.

The adaptive VI strategy modifies the voltage reference by embedding the virtual impedance, expressed as $\boldsymbol{Z}_v^{+(-)} = R_v^{+(-)} \pm jX_v^{+(-)}$, where

$$X_v^{+(-)} = n_{X/R}^{+(-)} R_v^{+(-)}$$
$$R_v^{+(-)} = \begin{cases} k_x^{+(-)}\left(\left|\boldsymbol{I}_{fdq}^{+(-)}\right| - I_{\text{th}}^{+(-)}\right), & \text{if } \left|\boldsymbol{I}_{fdq}^{+(-)}\right| > I_{\text{th}}^{+(-)} \\ 0, & \text{if } \left|\boldsymbol{I}_{fdq}^{+(-)}\right| \le I_{\text{th}}^{+(-)} \end{cases} \tag{1}$$

where $I_{\text{th}}^{+(-)}$ denotes the positive- (negative-) sequence current threshold that activates the adaptive VI. The parameter $k_x^{+(-)}$ is the positive- (negative-) sequence proportional gain. The user-specified parameter $n_{X/R}^{+(-)}$ is the positive- (negative-) sequence X/R ratio.

In contrast, the angle-priority CRS scheme directly readjusts the current reference, formulated as

$$\boldsymbol{I}_{fdq}^{\text{ref}+(-)} = \begin{cases} I_{\text{lim}}^{+(-)} e^{j\phi_I^{+(-)}}, & \text{if } \left|\boldsymbol{I}_{fdq}^{\text{cmd}+(-)}\right| > I_{\text{lim}}^{+(-)} \\ \boldsymbol{I}_{fdq}^{\text{cmd}+(-)}, & \text{if } \left|\boldsymbol{I}_{fdq}^{\text{cmd}+(-)}\right| \le I_{\text{lim}}^{+(-)} \end{cases} \tag{2}$$

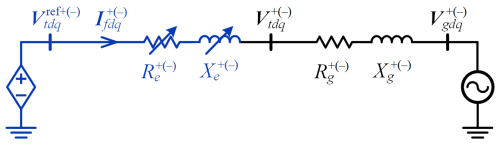


Fig. 2. Equivalent circuit of the GFM IBR when the CLC is activated.

where $I_{\text{lim}}^{+(-)}$ is the positive- (negative-) sequence current limit of the GFM IBR, and $\phi_I^{+(-)}$ is the user-defined phase angle of the positive- (negative-) sequence output current.

C. Clarifications and Assumptions

a) Theoretical analyses in the complex space vector and phasor domains are primarily applied to the steady state or quasi-steady state.
b) The dynamics of the current control loop is ignored, indicating $\boldsymbol{I}_{fdq}^{+(-)} \approx \boldsymbol{I}_{fdq}^{\text{ref}+(-)}$.
c) The voltage and current control dynamics are assumed to be significantly faster than the power control. Thus, the internal voltage $\boldsymbol{V}_{tdq}^{\text{ref}+(-)}$ is regarded as constant in the theoretical analysis.
d) The filter capacitance C_f, mainly used to filter high-frequency components of the output current, is omitted in the theoretical analysis.
e) The dynamics of the inductor and its series resistance (specifically, their related differentiate terms) are ignored in the theoretical analysis.

These conditions are introduced to facilitate the CLC design and transient stability analysis. All neglected factors in a)–e) are fully considered in the simulations and CHIL experiments presented in Section V.

D. Equivalent Circuit of the GFM IBR with CLC Activated

In a GFM IBR, the positive- (negative-) sequence internal voltage source $\boldsymbol{V}_{tdq}^{\text{ref}+(-)}$ is generated by the outer power control

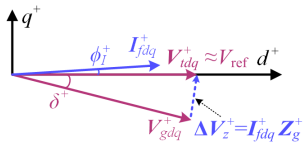


Fig. 3. Complex space vector diagram under normal operations.

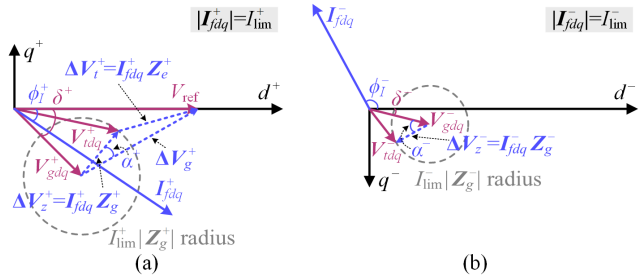


Fig. 4. Complex space vector diagram under asymmetric grid disturbances with CLC activated. (a) Positive-sequence. (b) Negative-sequence.

loops. This internal voltage is a virtual electromotive force embedded within the GFM IBR. Physically, the GFM IBR is connected to the external AC power grid via its terminal in both sequences. When a grid disturbance triggers the CLC of the GFM IBR, a virtual circuit can be always established by relating the internal voltage to the terminal voltage, regardless of the specific CLC implementation, as illustrated in Fig. 2. Based on the fundamental circuit laws, the voltage difference between the internal voltage $\boldsymbol{V}_{tdq}^{\text{ref}+(-)}$ and the terminal voltage $\boldsymbol{V}_{tdq}^{+(-)}$ can be represented in terms of the output current $\boldsymbol{I}_{fdq}^{+(-)}$ and an equivalent impedance $\boldsymbol{Z}_e^{+(-)}$. Accordingly, the impacts of different CLC schemes are characterised by $\boldsymbol{Z}_e^{+(-)}$, which can be derived using the GFM IBR's internal voltage, terminal voltage, and output current:

$$\boldsymbol{Z}_e^{+(-)} = R_e^{+(-)} \pm jX_e^{+(-)} = \frac{\boldsymbol{V}_{tdq}^{\text{ref}+(-)} - \boldsymbol{V}_{tdq}^{+(-)}}{\boldsymbol{I}_{fdq}^{+(-)}} \tag{3}$$

Unlike the equivalent circuit models in [33] and [34], which are developed under restrictive assumptions for the control system, such as negligible integral gains for the voltage control, or absence of additional virtual impedance, the model in (3) is established directly from observable variables, including $\boldsymbol{V}_{tdq}^{\text{ref}+(-)}$, $\boldsymbol{V}_{tdq}^{+(-)}$, and $\boldsymbol{I}_{fdq}^{+(-)}$. As such, it is not tied to any specific voltage control implementation or CLC scheme. The formulation in (3) is universally applicable to various voltage control strategies (VVC, ICC, or VAC) and CLC methods (VI or CRS), as their influences are inherently reflected in the terminal voltage and output current.

The general model in (3) is adopted as it characterises both VI- and CRS-based CLC effects as current- and voltage-dependent virtual impedances. This abstraction facilitates the development of the unified CLC design guideline in Section III and ensures the applicability of the guideline to both CLC types, as will be explained in Section IV.

III. Effect of CLC on GFM Capability Under Asymmetric Grid Disturbances

Under normal operating conditions, the CLC of the GFM IBR remains inactive, ensuring that the terminal voltage matches the internal voltage. Consequently, the complex space vectors $\boldsymbol{V}_{tdq}^{\text{ref}+(-)}$, $\boldsymbol{V}_{tdq}^{+(-)}$, and $\boldsymbol{V}_{gdq}^{+(-)}$ are shown in Fig. 3 within the positive- (negative-) sequence controller dq-frame, where $\boldsymbol{V}_{tdq}^{\text{ref}-} = \boldsymbol{V}_{tdq}^{-} = \boldsymbol{V}_{gdq}^{-} = 0$.

Under symmetric grid disturbances, where only positive-sequence components are present, the impact of CLC on the GFM capability has been demonstrated in [28]. For general asymmetric grid disturbances, including voltage dips and phase angle jumps that activate the CLC, the positive-sequence components propagate according to the symmetric conditions. Therefore, theoretical analysis in [28] remains applicable to the complex space vector relationships in the positive sequence, as illustrated in Fig. 4(a). The present work extends this analysis to the negative-sequence vectors, developed in Fig. 4(b).

Although Fig. 4(a) and Fig. 4(b) appear structurally similar, they are represented separately due to different setpoints for the positive- and negative-sequence internal voltage references $\boldsymbol{V}_{tdq}^{\text{ref}+}$ and $\boldsymbol{V}_{tdq}^{\text{ref}-}$. It is common practice to set $\boldsymbol{V}_{tdq}^{\text{ref}+} = V_{\text{ref}} + j0$ and $\boldsymbol{V}_{tdq}^{\text{ref}-} = 0 + j0$, thereby enabling the GFM IBR to obtain an internal equivalent circuit similar to that of a synchronous generator [25], [35]. Accordingly, the fictitious voltage drop across $\boldsymbol{Z}_e^+$ is represented by the voltage difference $\Delta\boldsymbol{V}_t^+$ between $\boldsymbol{V}_{tdq}^{\text{ref}+}$ and $\boldsymbol{V}_{tdq}^+$, as shown in Fig. 4(a). Similarly, the voltage drop across $\boldsymbol{Z}_e^-$ corresponds to $-\boldsymbol{V}_{tdq}^-$ due to the null vector $\boldsymbol{V}_{tdq}^{\text{ref}-}$, as depicted in Fig. 4(b). The voltage difference $\Delta\boldsymbol{V}_z^{+(-)}$ between $\boldsymbol{V}_{tdq}^{+(-)}$ and $\boldsymbol{V}_{gdq}^{+(-)}$ represents the voltage drop across $\boldsymbol{Z}_g^{+(-)}$, while $\Delta\boldsymbol{V}_g^+$ denotes the voltage difference between $\boldsymbol{V}_{tdq}^{\text{ref}+}$ and $\boldsymbol{V}_{gdq}^+$.

Under steady-state or quasi-steady-state conditions applicable to Fig. 4, the current vector $\boldsymbol{I}_{fdq}^{+(-)}$ is constrained by the CLC. Based on (1) and (2), the condition $\left|\boldsymbol{I}_{fdq}^{+(-)}\right| = I_{\text{lim}}^{+(-)}$ is satisfied for the VI-based CLC when $k_x^{+(-)}$ and $I_{\text{th}}^{+(-)}$ are properly tuned [23], and always holds for the CRS-based CLC [24], as will be explained in Section IV.B. The internal voltage reference $\boldsymbol{V}_{tdq}^{\text{ref}+(-)}$ is constant as indicated in Section II.C. In addition, $\boldsymbol{V}_{gdq}^{+(-)}$ and $I_{\text{lim}}^{+(-)}$ are fixed for a given disturbance and grid impedance [20]–[22]. Consequently, both $\Delta\boldsymbol{V}_g^+$ and $\left|\boldsymbol{I}_{fdq}^{+(-)}\right|$ remain constant, leading to a fixed $\left|\Delta\boldsymbol{V}_z^{+(-)}\right|$. However, the phase angle of $\boldsymbol{I}_{fdq}^{+(-)}$ may vary depending on the CLC strategy, thereby affecting $\boldsymbol{V}_{tdq}^{+(-)}$ and resulting in different GFM capabilities.

As mandated by multiple system operators [5]–[8], during severe asymmetric disturbances, the GFM IBR is required to maintain a balanced terminal voltage to the maximum extent within its physical limits for GFM capability enhancement. Therefore, when the CLC is active, the positive-sequence terminal voltage $\boldsymbol{V}_{tdq}^+$ should be maximised while the negative-sequence terminal voltage $\boldsymbol{V}_{tdq}^-$ should be minimised. In other words, $\boldsymbol{V}_{tdq}^{+(-)}$ should closely align with $\boldsymbol{V}_{tdq}^{\text{ref}+(-)}$ to minimise variations in terminal voltage by reducing $\Delta\boldsymbol{V}_t^+$ and $\boldsymbol{V}_{tdq}^-$, thereby enhancing GFM capability.

Referring to Fig. 4, $\left|\Delta \boldsymbol{V}_t^+\right|$ and $\left|\boldsymbol{V}_{tdq}^-\right|$ are derived as

$$\left|\Delta \boldsymbol{V}_t^+\right| = \sqrt{\left|\Delta \boldsymbol{V}_g^+\right|^2 + \left|\Delta \boldsymbol{V}_z^+\right|^2 - 2\left|\Delta \boldsymbol{V}_g^+\right|\left|\Delta \boldsymbol{V}_z^+\right|\cos\alpha^+} \tag{4}$$

$$\left|\boldsymbol{V}_{tdq}^-\right| = \sqrt{\left|\boldsymbol{V}_{gdq}^-\right|^2 + \left|\Delta \boldsymbol{V}_z^-\right|^2 - 2\left|\boldsymbol{V}_{gdq}^-\right|\left|\Delta \boldsymbol{V}_z^-\right|\cos\alpha^-} \tag{5}$$

where α^+ denotes the phase angle difference between $\Delta \boldsymbol{V}_g^+$ and $\Delta \boldsymbol{V}_z^+$, and α^- denotes that between $-\boldsymbol{V}_{gdq}^-$ and $\Delta \boldsymbol{V}_z^-$.

Since $\left|\Delta \boldsymbol{V}_g^+\right|$ and $\left|\Delta \boldsymbol{V}_z^+\right|$ are constant values as previously stated, $\left|\Delta \boldsymbol{V}_t^+\right|$ in (4) decreases as α^+ decreases. Likewise, the constants $\left|\boldsymbol{V}_{gdq}^-\right|$, $\left|\Delta \boldsymbol{V}_z^-\right|$, along with a smaller α^-, lead to a reduced $\left|\boldsymbol{V}_{tdq}^-\right|$ in (5). Therefore, $\left|\Delta \boldsymbol{V}_t^+\right|$ and $\left|\boldsymbol{V}_{tdq}^-\right|$ reach their minimum values at $\alpha^+ = 0$ and $\alpha^- = 0$. This establishes a collinear relationship between $\Delta \boldsymbol{V}_t^+$ and $\Delta \boldsymbol{V}_z^+$ in the positive sequence ($\alpha^+ = 0$), and between $\boldsymbol{V}_{tdq}^-$ and $\Delta \boldsymbol{V}_z^-$ in the negative sequence ($\alpha^- = 0$).

Drawing upon Fig. 2 and Fig. 4, $\alpha^{+(-)} = 0$ further implies the desired phase angle of $\boldsymbol{Z}_e^{+(-)}$:

$$\angle \boldsymbol{Z}_e^{+(-)} = \angle \boldsymbol{Z}_g^{+(-)} \Rightarrow X_e^{+(-)}/R_e^{+(-)} = X_g^{+(-)}/R_g^{+(-)} \tag{6}$$

Substituting $\alpha^{+(-)} = 0$ into (4) and (5) obtains $\left|\Delta \boldsymbol{V}_t^+\right|_{\min}$ and $\left|\boldsymbol{V}_{tdq}^-\right|_{\min}$, formulated as

$$\left|\Delta \boldsymbol{V}_t^+\right|_{\min} = \left|\left|\Delta \boldsymbol{V}_g^+\right| - \left|\Delta \boldsymbol{V}_z^+\right|\right| \tag{7}$$

$$\left|\boldsymbol{V}_{tdq}^-\right|_{\min} = \left|\left|\boldsymbol{V}_{gdq}^-\right| - \left|\Delta \boldsymbol{V}_z^-\right|\right| \tag{8}$$

The above analysis establishes a generalised design guideline for enhancing the GFM capability of IBRs under current limitation: Aligning the phase angle of the positive- (negative-) sequence equivalent impedance $\boldsymbol{Z}_e^{+(-)}$ emulated by the CLC with that of the positive- (negative-) sequence grid impedance $\boldsymbol{Z}_g^{+(-)}$. The positive-sequence formulations in (4) and (7) have been previously reported in [28]. Thus, the main contribution here lies in extending the guideline to the negative-sequence domain, as developed in (5) and (8). This extension addresses an existing gap in [28], where the negative-sequence impedance phase angle $\angle \boldsymbol{Z}_e^-$ is not considered. Moreover, the extended guideline is consistent with the simulation-based findings in [27], while offering analytical insights into the link between the terminal voltage and the phase angle of the virtual impedance emulated by the CLC in both sequences.

Notably, the desired negative-sequence phase angle $\angle \boldsymbol{Z}_e^-$ in (6) is derived under the condition that $\boldsymbol{V}_{tdq}^{\text{ref}-} = 0 + j0$. Due to the control flexibility of the GFM IBR, a nonzero $\boldsymbol{V}_{tdq}^{\text{ref}-}$ can also be assigned. In this case, $\angle \boldsymbol{Z}_e^-$ no longer satisfies (6) and needs to be rederived by extending Fig. 4(b) to incorporate the nonzero $\boldsymbol{V}_{tdq}^{\text{ref}-}$ vector. By properly designing $\angle \boldsymbol{Z}_e^-$ based on the assigned nonzero $\boldsymbol{V}_{tdq}^{\text{ref}-}$, the negative-sequence terminal voltage can still be minimised, as in the case of $\boldsymbol{V}_{tdq}^{\text{ref}-} = 0 + j0$ where $\angle \boldsymbol{Z}_e^-$ satisfies (6). Different $\boldsymbol{V}_{tdq}^{\text{ref}-}$ and $\angle \boldsymbol{Z}_e^-$ settings merely provide alternative representations of the negative-sequence internal voltage and equivalent impedance, which are fictitious and reflect different CLC implementations. As long as the control objective remains the same, namely, minimising the negative-sequence terminal voltage, the resulting negative-sequence terminal voltage and output current remain identical. However, setting $\boldsymbol{V}_{tdq}^{\text{ref}-} = 0 + j0$ during asymmetric faults offers two advantages. It preserves a balanced internal voltage for the GFM IBR and simplifies control implementation, since $\angle \boldsymbol{Z}_e^-$ can then be directly set equal to $\angle \boldsymbol{Z}_g^-$ without involving a nonzero $\boldsymbol{V}_{tdq}^{\text{ref}-}$.

IV. Proposed Current-Limiting Strategy

A. Improved CLC Schemes for GFM Capability Enhancement

To enhance IBR's GFM capability during current limitation, the CLC should reshape its emulated impedance $\boldsymbol{Z}_e^{+(-)}$ to satisfy (6). By invoking (1) and (2), $n_{X/R}^{+(-)}$ in the adaptive VI scheme and $\phi_I^{+(-)}$ in the angle-priority CRS scheme provide control degrees of freedom that enable $\boldsymbol{Z}_e^{+(-)}$ to be tailored to the desired properties.

1) VI-Based Scheme: According to Fig. 1, when the VI is active, it explicitly manipulates the voltage difference between $\boldsymbol{V}_{tdq}^{\text{ref}+(-)}$ and $\boldsymbol{V}_{tdq}^{+(-)}$, indicating $\boldsymbol{Z}_e^{+(-)} = \boldsymbol{Z}_v^{+(-)}$. Thus, $n_{X/R}^{+(-)}$ in (1) can be directly set to $X_g^{+(-)}/R_g^{+(-)}$ to meet (6).

2) CRS-Based Scheme: The CRS-based CLC is embedded between the voltage control and current control, as shown in Fig. 1. In this case, the angle-priority CRS method implicitly adjusts the emulated impedance $\boldsymbol{Z}_e^{+(-)}$ by modifying the current reference [see (2)]. Thus, establishing an intuitive mathematical relationship between $\phi_I^{+(-)}$ and $\boldsymbol{Z}_e^{+(-)}$ to satisfy (6) is challenging. Recalling (3), $\boldsymbol{Z}_e^{+(-)}$ can be expressed in terms of the vectors $\boldsymbol{V}_{tdq}^{\text{ref}+(-)}$, $\boldsymbol{V}_{tdq}^{+(-)}$, and $\boldsymbol{I}_{fdq}^{+(-)}$. When CLC is triggered, these vectors are given by

$$\begin{cases} \boldsymbol{V}_{tdq}^{\text{ref}+} = V_{\text{ref}} + j0, \boldsymbol{V}_{tdq}^{\text{ref}-} = 0 + j0 \\ \boldsymbol{V}_{tdq}^{+(-)} = \boldsymbol{I}_{fdq}^{+(-)}\boldsymbol{Z}_g^{+(-)} + V_g^{+(-)}e^{j\delta^{+(-)}} \\ \boldsymbol{I}_{fdq}^{+(-)} = I_{\lim}^{+(-)}e^{j\phi_I^{+(-)}} \end{cases} \tag{9}$$

where $V_g^{+(-)}$ and $\delta^{+(-)}$ denote the magnitude and phase angle of $\boldsymbol{V}_{gdq}^{+(-)}$, respectively.

Plugging (9) into (3) results in

$$\boldsymbol{Z}_e^{+(-)} = R_e^{+(-)} \pm jX_e^{+(-)}$$

$$\begin{cases} R_e^+ = \frac{V_{\text{ref}}\cos\left(\phi_I^+\right) - V_g^+\cos\left(\delta^+ - \phi_I^+\right)}{I_{\lim}^+} - R_g^+ \\ X_e^+ = \frac{-V_{\text{ref}}\sin\left(\phi_I^+\right) - V_g^+\sin\left(\delta^+ - \phi_I^+\right)}{I_{\lim}^+} - X_g^+ \end{cases} \tag{10}$$

$$\begin{cases} R_e^- = \frac{-V_g^-\cos\left(\delta^- - \phi_I^-\right)}{I_{\lim}^-} - R_g^- \\ X_e^- = \frac{V_g^-\sin\left(\delta^- - \phi_I^-\right)}{I_{\lim}^-} - X_g^- \end{cases}$$

By substituting (10) into (6) and performing cross-multiplication [define $\angle \boldsymbol{Z}_g^{+(-)} = \arctan(X_g^{+(-)}/R_g^{+(-)}) = \phi_{z_g}^{+(-)}$], the following conditions are derived:

$$V_g^+\sin\left(\phi_I^+ + \phi_{z_g}^+ - \delta^+\right) = V_{\text{ref}}\sin\left(\phi_I^+ + \phi_{z_g}^+\right) \tag{11}$$

$$V_g^-\sin\left(\phi_I^- - \phi_{z_g}^- - \delta^-\right) = 0 \tag{12}$$

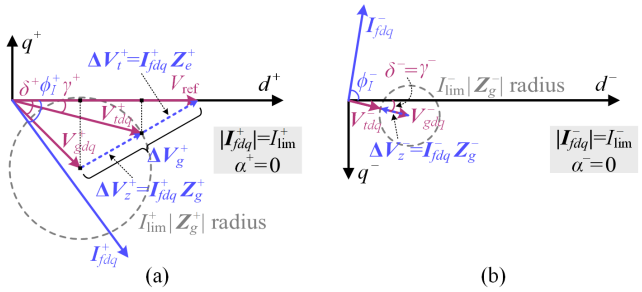


Fig. 5. Complex space vector diagram under asymmetric grid disturbances when $\alpha^{+(-)}=0$. (a) Positive-sequence. (b) Negative-sequence.

Expanding the left-hand side of (11) allows the derivation of $\tan(\phi_I^+ + \phi_{z_g}^+)$, from which ϕ_I^+ is obtained as

$$\phi_I^+ = -\phi_{z_g}^+ + \arctan\frac{-V_g^+\sin(\delta^+)}{V_{\text{ref}} - V_g^+\cos(\delta^+)} \tag{13}$$

Similarly, ϕ_I^- is directly derived from (12) as

$$\phi_I^- = \phi_{z_g}^- + \delta^- - \pi \tag{14}$$

With (13) and (14), the design principle in (6) is extended to the angle-priority CRS method. However, V_g^+ and $\delta^{+(-)}$ are system-wide variables that are not readily measurable in practice. Thus, further simplifications of (13) and (14) are required. As demonstrated in Section III, when (6) holds, $\alpha^{+(-)}=0$. Consequently, vector relationships in Fig. 4 can be reformulated as depicted in Fig. 5.

For the positive-sequence components in Fig. 5(a), projecting the endpoints of $\boldsymbol{V}_{gdq}^+$ and $\boldsymbol{V}_{tdq}^+$ onto the d^+-axis yields the following relationship:

$$\frac{V_g^+\sin(\delta^+)}{V_{\text{ref}} - V_g^+\cos(\delta^+)} = \frac{V_t^+\sin(\gamma^+)}{V_{\text{ref}} - V_t^+\cos(\gamma^+)} = \frac{V_{tq}^+}{V_{\text{ref}} - V_{td}^+} \tag{15}$$

where V_t^+ and γ^+ represent the magnitude and phase angle of $\boldsymbol{V}_{tdq}^+$, respectively.

Likewise, for negative-sequence components in Fig. 5(b), the phase angle relationship can be expressed as

$$\delta^- = \gamma^- = \begin{cases} \arctan\frac{V_{tq}^-}{V_{td}^-}, & \text{if } V_{td}^- \geq 0 \\ \arctan\frac{V_{tq}^-}{V_{td}^-} \pm \pi, & \text{if } V_{td}^- < 0 \end{cases} \tag{16}$$

where γ^- denotes the phase angle of $\boldsymbol{V}_{tdq}^-$.

By substituting (15) and (16) into (13) and (14), respectively, the system-wide variables V_g^+ and $\delta^{+(-)}$ are replaced with local variables $V_{td}^{+(-)}$ and $V_{tq}^{+(-)}$, yielding:

$$\phi_I^+ = -\phi_z^+ + \arctan\frac{-V_{tq}^+}{V_{\text{ref}} - V_{td}^+} \tag{17}$$

$$\phi_I^- = \phi_z^- + \begin{cases} \arctan\frac{V_{tq}^-}{V_{td}^-} - \pi, & \text{if } V_{td}^- \geq 0 \\ \arctan\frac{V_{tq}^-}{V_{td}^-}, & \text{if } V_{td}^- < 0 \end{cases} \tag{18}$$

where $\phi_z^{+(-)}$ is the phase angle of the equivalent impedance emulated by the angle-priority CRS-based CLC. Ideally, $\phi_z^{+(-)}$ should match $\phi_{z_g}^{+(-)}$ to enhance IBR's GFM capability.

The equivalent transformations from the system-wide variables V_g^+ and $\delta^{+(-)}$ to the local variables $V_{td}^{+(-)}$ and $V_{tq}^{+(-)}$ are derived from the inherent voltage vector relationships shown in Fig. 5, which are always valid for (13) and (14). These vector relationships reflect the aligned phase angles between the CLC-emulated impedance and the grid impedance, without imposing any constraints on the magnitude of the grid impedance. Therefore, regardless of the grid strength, the simplifications involving V_g^+ and $\delta^{+(-)}$ remain applicable. Even under extremely weak grid conditions, where $\delta^{+(-)}$ may become significant, the variables V_g^+ and $\delta^{+(-)}$ in (13) and (14) can still be equivalently transformed to $V_{td}^{+(-)}$ and $V_{tq}^{+(-)}$ as used in (17) and (18).

The theoretical analysis above demonstrates that ensuring $\phi_I^{+(-)}$ conforms to (17) and (18) enables the emulated impedance of the angle-priority CRS method to satisfy (6). Consequently, the angle-priority CRS-based CLC can also enhance GFM capability.

Assuming $\phi_z^{+(-)} = \arctan(X_e^{+(-)}/R_e^{+(-)})$ and substituting $\phi_I^{+(-)} = \arctan(I_{fq}^{\text{ref}+(-)}/I_{fd}^{\text{ref}+(-)})$ into (17) and (18), the resulting $I_{fq}^{\text{ref}+(-)}/I_{fd}^{\text{ref}+(-)}$ is equivalent to that generated by the static VAC method [24], [36]. This shows that, the proposed CRS-based CLC replicates the control behaviour of the VAC formulation. Accordingly, GFM IBRs employing the proposed CRS-based scheme retain the voltage source characteristics. As in [24] and [36], applying a low-pass filter (LPF) with a time constant of 5–10 ms to the voltage feedback ensures adequate damping and prevents potential control interactions. Thus, the same LPF is incorporated into the proposed CRS-based framework in (17) and (18).

Additionally, the proposed CLC methods operate within the inner voltage and current control loops and primarily reshape the equivalent impedance of the GFM IBR during severe disturbances. In contrast, alternative CLC approaches that are designed for the outer power control loops, such as the active power reference limiter [24], virtual power feedback [37], and hybrid-synchronisation control [38], mainly regulate the magnitude and phase angle of the internal voltage source. Such methods lie beyond the scope of this paper. But it should be noted that the proposed VI and CRS methods can be flexibly combined with CLC schemes tailored for the power control loops to simultaneously manipulate both the internal voltage source and the equivalent impedance during fault conditions, thereby further enhancing transient stability performance.

Remark 1. Comparison and selection between both implementations: In practice, the VI-based CLC is usually restricted by slow dynamic response due to the limited bandwidth of the voltage control loop [23]. Its implementation also requires multiloop interaction management and careful parameter tuning [see Section IV.B]. Given these limitations, the CRS-based CLC is adopted as the primary solution in this study due to its faster dynamic response, enhanced current-limiting capability, and minimal parameter tuning requirements [10].

Remark 2. Coordination with conventional inverter overcurrent protection: Due to the intrinsic response time of the IBR control system, the proposed CLC limits the output current within the first cycle following a disturbance. This behaviour differs from the conventional, faster-acting inverter protections, such as the gate-level overcurrent protection, which

operates on a microsecond timescale by immediately freezing PWM signals once a predefined threshold is exceeded. To ensure reliable and safe operation of the GFM IBR, effective coordination is required between the proposed CLC (control-level limiting) and conventional inverter overcurrent protection (hardware-level protection). This includes: i) threshold coordination, the current limit enforced by the CLC must remain below the trigger threshold of the inverter protection. For example, the CLC may cap the output current at 1.5 p.u., preventing it from reaching the hardware protection threshold of 2.0 p.u.; ii) timing coordination, a clear timing hierarchy between the millisecond-scale response of the CLC and the microsecond-scale reaction of inverter protection ensures that the CLC has adequate time to respond, thereby avoiding unnecessary inverter shutdown.

Remark 3. Applicability in multi-inverter systems: In a multi-inverter system, the proposed CLC scheme can be independently implemented on each IBR, as their current-limiting function mainly relies on local variables including output current and terminal voltage. However, due to interactions among multiple units, accurately obtaining the phase angle of the grid impedance for each IBR can be challenging. This may degrade the performance of the proposed CLC. Nevertheless, as will be discussed in Section IV.D, the proposed method is robust even when the available grid impedance angle deviates significantly from its actual value. Hence, the CLC scheme remains effective in multi-inverter configurations.

B. Compliance with Modern GCs

Before implementing the proposed CLC scheme in GFM IBRs, a rigorous evaluation is necessary to ensure compliance with recent GCs, such as [20]–[22], as outlined below.

1) Maximum Utilisation of IBR's Current Generation Capacity: During severe disturbances, maximising current injection at the fundamental frequency from IBRs is crucial for maintaining system stability [13], and is therefore a mandatory requirement in modern GCs [20]–[22]. This necessitates that, during asymmetric disturbances, the output current in at least one of the three phases reaches the IBR's limit. To satisfy this requirement, $I_{\rm lim}^{+(-)}$ needs to be properly designed within the CLC. Specifically, the maximum phase current of the GFM IBR can be determined from $\boldsymbol{I}_{fdq}^{+(-)}$ as [23], [24], [26]

$$I_{\rm pha,max} = \sqrt{\left|\boldsymbol{I}_{fdq}^{+}\right|^2 + \left|\boldsymbol{I}_{fdq}^{-}\right|^2 + 2I_{\Delta}} \tag{19}$$

where $I_{\Delta} = \max_{\varphi}\left\{\left|\boldsymbol{I}_{fdq}^{+}\right|\left|\boldsymbol{I}_{fdq}^{-}\right|\cos\left(\angle\boldsymbol{I}_{fdq}^{+} + \angle\boldsymbol{I}_{fdq}^{-} + \varphi\right)\right\}$, with $\varphi \in [0, -2\pi/3, 2\pi/3]$.

To prevent thermal damage of the inverter, $I_{\rm pha,max}$ must not exceed the maximum permissible current of the GFM IBR, denoted as $I_{\rm lim}$, which typically ranges from 1.1 to 2 p.u. Imposing the constraint $I_{\rm pha,max} \le I_{\rm lim}$ in (19) yields:

$$\left|\boldsymbol{I}_{fdq}^{+}\right| \le \sqrt{I_{\rm lim}^2 - \left|\boldsymbol{I}_{fdq}^{-}\right|^2 - 2I_{\Delta}} \stackrel{\rm def}{=} I_{\rm lim}^{+} \tag{20}$$

Furthermore, $I_{\rm lim}^{-}$ can be determined using the widely adopted K-factor method, as specified in [21]–[24]:

$$I_{\rm lim}^{-} = K^{-}\left|\boldsymbol{V}_{tdq}^{-}\right| \tag{21}$$

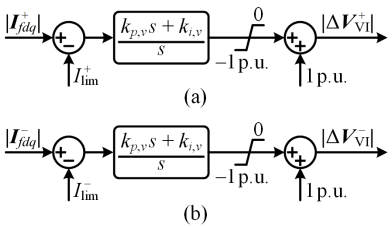


Fig. 6. PI control loop applied to derive the voltage drop across the virtual impedance. (a) $\left|\Delta\boldsymbol{V}_{\rm VI}^{+}\right|$. (b) $\left|\Delta\boldsymbol{V}_{\rm VI}^{-}\right|$.

where K^{-} typically ranges from 2 to 6.

For the proposed CRS-based CLC scheme, by respectively configuring $I_{\rm lim}^{+}$ and $I_{\rm lim}^{-}$ according to (20) and (21), the current generation capacity of the inverter can be fully utilised under current-limiting conditions, ensuring compliance with GCs. In contrast, the VI-based scheme requires additional parameter coordination. Beyond aligning $I_{\rm lim}^{+(-)}$ with (20) and (21), appropriate settings for the current threshold $I_{\rm th}^{+(-)}$ and the proportional gain $k_x^{+(-)}$ are essential. Specifically, the positive-sequence current threshold $I_{\rm th}^{+}$ needs to be set below $I_{\rm lim}^{+}$ to ensure that the positive-sequence VI operates before a positive-sequence overcurrent occurs. This yields:

$$I_{\rm th}^{+} = I_{\rm lim}^{+} - \Delta I_{\rm tri}^{+} \tag{22}$$

where $I_{\rm tri}^{+}$ is the positive-sequence trigger margin, typically selected within 1 p.u. [28], [39].

For the negative-sequence current threshold $I_{\rm th}^{-}$, a small fixed value is required to satisfy two conditions: first, that the negative-sequence adaptive VI remains inactive under symmetric fault conditions (i.e., $I_{\rm th}^{-} > 0$); and second, that it is triggered before the onset of negative-sequence overcurrent (i.e., $I_{\rm th}^{-} < I_{\rm lim}^{-}$). Assuming the negative-sequence VI is activated only when the negative-sequence terminal voltage magnitude $\left|\boldsymbol{V}_{tdq}^{-}\right|$ exceeds a predefined threshold $\Delta V_{\rm tri}^{-} \in [0, 0.1]$ p.u. [21], [23], the threshold $I_{\rm th}^{-}$ can be determined accordingly. For example, with $\Delta V_{\rm tri}^{-} = 0.05$ p.u., (21) gives $I_{\rm lim}^{-} = 0.1$ p.u. for a minimum $K^{-} = 2$ and $\left|\boldsymbol{V}_{tdq}^{-}\right| = 0.05$ p.u. Thus, $I_{\rm th}^{-}$ should be set in the range of 0–0.1 p.u.

With $I_{\rm lim}^{+(-)}$ and $I_{\rm th}^{+(-)}$ defined, the remaining key parameter in the VI-based method is $k_x^{+(-)}$, which determines the ability of VI to limit phase current strictly to $I_{\rm lim}$.

During asymmetric grid disturbances, the voltage drop across the positive- (negative-) sequence virtual impedance can be defined as $\left|\Delta\boldsymbol{V}_{\rm VI}^{+(-)}\right|$. Accordingly, $k_x^{+(-)}$ should satisfy [12], [37]:

$$k_x^{+(-)} \ge \frac{\left|\Delta\boldsymbol{V}_{\rm VI}^{+(-)}\right|}{I_{\rm lim}^{+(-)}\sqrt{1 + n_{X/R}^{+(-)2}}\left(I_{\rm lim}^{+(-)} - I_{\rm th}^{+(-)}\right)} \tag{23}$$

Assigning a fixed value to $\left|\Delta\boldsymbol{V}_{\rm VI}^{+(-)}\right|$ under given fault conditions allows direct computation of a constant $k_x^{+(-)}$, following the classical tuning approach [39]. However, using a constant $k_x^{+(-)}$ causes the steady-state fault current to settle between $I_{\rm th}^{+(-)}$ and $I_{\rm lim}^{+(-)}$ [13], [37], leading to underutilised current headroom and potential GC violations. To address this

limitation, $k_x^{+(-)}$ needs to be adaptively adjusted depending on the fault conditions. As documented in [23], a separate proportional-integral (PI) control loop can be applied to dynamically correct the ongoing voltage drop $\left|\Delta \boldsymbol{V}_{\text{VI}}^{+}\right|$ across the positive-sequence virtual impedance, as shown in Fig. 6(a). This loop operates during current limitation and adjusts $\left|\Delta \boldsymbol{V}_{\text{VI}}^{+}\right|$ to generate a proper k_x^{+} such that $\left|\boldsymbol{I}_{fdq}^{+}\right| = I_{\text{lim}}^{+}$.

Similarly, a PI control loop in Fig. 6(b) is employed to regulate the negative-sequence voltage drop $\left|\Delta \boldsymbol{V}_{\text{VI}}^{-}\right|$ during asymmetric disturbances, thereby generating an appropriate k_x^{-} that ensures $\left|\boldsymbol{I}_{fdq}^{-}\right| = I_{\text{lim}}^{-}$.

Once $\left|\Delta \boldsymbol{V}_{\text{VI}}^{+}\right|$ and $\left|\Delta \boldsymbol{V}_{\text{VI}}^{-}\right|$ are obtained from Fig. 6, the corresponding k_x^{+} and k_x^{-} values can be calculated using (23) by converting the inequality into an equality. With the complete parameter set $I_{\text{lim}}^{+(-)}$, $I_{\text{th}}^{+(-)}$, and $k_x^{+(-)}$ determined, the inverter's current generation capacity can be fully exploited under the proposed VI-based CLC scheme.

2) Positive-Sequence Reactive Current: In transmission systems, prioritising the positive-sequence reactive current I_Q^{+} is essential for maintaining voltage stability during disturbances [20]–[22]. Recent GCs specify the minimum required $\left|I_Q^{+}\right|$ under disturbances based on the retained positive-sequence terminal voltage [20], [21]. Hence, $\left|I_Q^{+}\right|$ supplied by the proposed CLC needs to be analysed for compliance.

The proposed VI- and CRS-based CLC schemes are building upon the design guideline in (6). Accordingly, the phase angle of the positive-sequence current $\boldsymbol{I}_{fdq}^{+}$ generated by both schemes satisfies (17). In transmission networks, where $\angle \boldsymbol{Z}_g^{+} \approx \pi/2$ [28], substituting $\phi_z^{+} \approx \pi/2$ into (17) yields:

$$\tan \phi_I^{+} = \frac{I_{fq}^{+}}{I_{fd}^{+}} = \frac{V_{\text{ref}} - V_{td}^{+}}{V_{tq}^{+}} \tag{24}$$

Further simplification of (24) leads to

$$I_{fd}^{+} = \frac{V_{td}^{+} I_{fd}^{+} + V_{tq}^{+} I_{fq}^{+}}{V_{\text{ref}}}, \quad I_{fq}^{+} = \frac{V_{\text{ref}} I_{fd}^{+} - V_{td}^{+} I_{fd}^{+}}{V_{tq}^{+}} \tag{25}$$

The positive-sequence active current I_P^{+} and reactive current I_Q^{+} can be written as [23]

$$I_P^{+} = \frac{V_{td}^{+} I_{fd}^{+} + V_{tq}^{+} I_{fq}^{+}}{\left|\boldsymbol{V}_{tdq}^{+}\right|}, \quad I_Q^{+} = \frac{-V_{td}^{+} I_{fq}^{+} + V_{tq}^{+} I_{fd}^{+}}{\left|\boldsymbol{V}_{tdq}^{+}\right|} \tag{26}$$

Comparing I_{fd}^{+} in (25) with I_P^{+} in (26) obtains $I_{fd}^{+} = I_P^{+}\left|\boldsymbol{V}_{tdq}^{+}\right|/V_{\text{ref}}$. As per the latest specifications in [5]–[9], the GFM IBR should maintain a nearly constant internal voltage phasor during transients, which implies $V_{\text{ref}} \approx 1$ p.u. Thus, $I_{fd}^{+} \approx I_P^{+}\left|\boldsymbol{V}_{tdq}^{+}\right|$. Substituting this condition along with I_{fq}^{+} from (25) into I_Q^{+} from (26) gives:

$$I_Q^{+} = \frac{\left|\boldsymbol{V}_{tdq}^{+}\right|^2 - V_{td}^{+}}{V_{tq}^{+}} I_P^{+} \tag{27}$$

When the CLC is active, the following constraint applies:

$$I_{\text{lim}}^{+} = \sqrt{I_P^{+2} + I_Q^{+2}} \tag{28}$$

It follows from (27) to (28) that:

$$I_Q^{+2} = \frac{\left(\left|\boldsymbol{V}_{tdq}^{+}\right|^2 - V_{td}^{+}\right)^2}{\left|\boldsymbol{V}_{tdq}^{+}\right|^2 \left(\left|\boldsymbol{V}_{tdq}^{+}\right|^2 - 2V_{td}^{+} + 1\right)} I_{\text{lim}}^{+2} \tag{29}$$

Since the positive-sequence terminal voltage vectors are design to be aligned with the d^{+}-axis, as illustrated in Fig. 1, $V_{td}^{+} \approx \left|\boldsymbol{V}_{tdq}^{+}\right|$ holds under voltage sag conditions [11], [26]. Consequently, I_Q^{+} in (29) is simplified to

$$\left|I_Q^{+}\right| \approx I_{\text{lim}}^{+} \tag{30}$$

The above derivation reveals that, when the proposed CLC is activated, $\left|I_Q^{+}\right|$ can closely approach I_{lim}^{+}, provided $\phi_z^{+} = \angle \boldsymbol{Z}_g^{+} \approx \pi/2$. When $\angle \boldsymbol{Z}_g^{+}$ deviates moderately from $\pi/2$ (e.g., $\angle \boldsymbol{Z}_g^{+} = 70^\circ - 90^\circ$), which represents typical transmission scenarios, the condition in (30) still holds due to the predominantly inductive nature of transmission systems. Under these conditions, the proposed CLC is always able to provide a value of $\left|I_Q^{+}\right|$ that exceeds the minimum reactive current threshold required in [21], [22], thereby ensuring the GC compliance. When $\angle \boldsymbol{Z}_g^{+}$ is significantly lower, indicating the absence of transmission network conditions, the ability of the proposed CLC method to deliver positive-sequence reactive current increasingly depends on the X_g^{+}/R_g^{+} ratio, as reflected in (17). For example, $\phi_z^{+} = \angle \boldsymbol{Z}_g^{+} = 45^\circ$, corresponding to $X_g^{+}/R_g^{+} = 1$, results in the GFM IBR generating comparable levels of positive-sequence active and reactive currents under the proposed CLC scheme. This behaviour aligns with the dynamic voltage support requirements specified in [40] (designated for distribution systems), which suggest that the active and reactive current injection should depend on the X/R ratio of the area electric power systems.

3) Negative-Sequence Reactive Current: To guarantee the proper operation of protection elements and mitigate potential overvoltage risks, the phase angle of the negative-sequence current supplied by IBRs in transmission systems must lead the negative-sequence voltage by 90° to 100° [21], [22]. This requirement indicates that the negative-sequence current from GFM IBRs should be approximately inductive under asymmetric disturbances.

The components I_{fd}^{-} and I_{fq}^{-} can be expressed in terms of the negative-sequence active current I_P^{-}, reactive current I_Q^{-}, and $\boldsymbol{V}_{tdq}^{-} = V_{td}^{-} + jV_{tq}^{-}$ as follows [23]:

$$I_{fd}^{-} = \frac{V_{td}^{-} I_P^{-} + V_{tq}^{-} I_Q^{-}}{\left|\boldsymbol{V}_{tdq}^{-}\right|}, \quad I_{fq}^{-} = \frac{V_{tq}^{-} I_P^{-} - V_{td}^{-} I_Q^{-}}{\left|\boldsymbol{V}_{tdq}^{-}\right|} \tag{31}$$

For an inductive negative-sequence current, I_P^{-} needs to be zero, and I_Q^{-} should be negative. As a result, (31) becomes:

$$I_{fd}^{-} = \frac{-V_{tq}^{-}\left|I_Q^{-}\right|}{\left|\boldsymbol{V}_{tdq}^{-}\right|}, \quad I_{fq}^{-} = \frac{V_{td}^{-}\left|I_Q^{-}\right|}{\left|\boldsymbol{V}_{tdq}^{-}\right|} \tag{32}$$

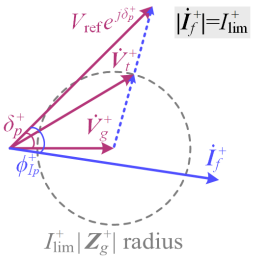


Fig. 7. Phasor diagram of the GFM IBR under the proposed CLC method.

Based on (32), the following relationship holds:

$$\frac{I_{fq}^-}{I_{fd}^-} = -\frac{V_{td}^-}{V_{tq}^-} \tag{33}$$

During asymmetric disturbances with the proposed CLC activated, the phase angle $\phi_I^- = I_{fq}^-/I_{fd}^-$ of the negative-sequence output current $\boldsymbol{I}_{fdq}^-$ satisfies (18). Assuming $\phi_z^- \approx \pi/2$ for the transmission system, (33) can be directly recovered from (18). This demonstrates that the proposed CLC scheme enables the GFM IBR to generate inductive negative-sequence current in transmission grids, thereby meeting the GC requirement of a 90° to 100° phase difference between the negative-sequence output current and terminal voltage.

The above discussion indicates that the proposed negative-sequence CLC operates in the negative-sequence voltage mitigation mode. Three other commonly adopted negative-sequence control modes include balanced current control, active power oscillation suppression, and reactive power oscillation suppression. However, these modes contribute minimally to maintaining a balanced terminal voltage during asymmetric faults. As a result, they fail to meet the requirements outlined in GFM-specific standards [5]–[9], thus, provide limited enhancement to GFM capability. In applications where one of these three alternative modes is necessary, it can be readily achieved using the flexible current command generation method in [37].

C. Transient Stability Analysis

To analyse the transient stability achieved by the proposed CLC methods, the power-angle relationships of both the proposed VI- and CRS-based schemes are established based on the positive-sequence components. The analysis is conducted in the phasor domain, rather than the complex space vector domain used previously, as the former is preferable for transient stability, as shown in Fig. 7. Although the layout of Fig. 5(a) and Fig. 7 is the same, the reference axis in Fig. 7 is rotated to align with the positive-sequence grid voltage phasor $\dot{\boldsymbol{V}}_g^+$. In this figure, both $\delta_p^+ > 0$ and $\phi_{Ip}^+ > 0$ hold.

1) Power-Angle Characterisation for VI-Based Scheme: When the proposed VI-based CLC method is activated, the following condition holds according to Fig. 7:

$$\left| \frac{V_{\text{ref}}e^{j\delta_p^+} - V_g^+ e^{j0}}{R_v^+ + R_g^+ + j\left(X_v^+ + X_g^+\right)} \right| = I_{\text{lim}}^+ \tag{34}$$

where R_v^+ and X_v^+ can be determined by substituting $\left|\boldsymbol{I}_{fdq}^+\right| = I_{\text{lim}}^+$ into (1).

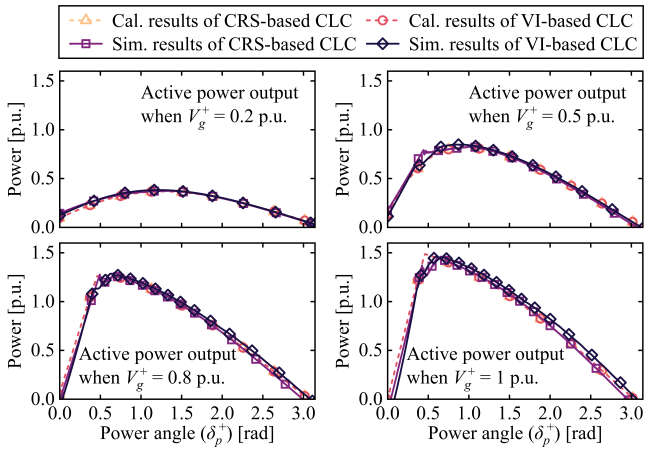


Fig. 8. Power-angle curves under different grid voltage magnitudes.

For a given grid voltage magnitude V_g^+ and power angle δ_p^+, k_x^+ needs to be first calculated using (34). If no valid solution exists or the computed value of k_x^+ is negative, the adaptive VI is inactive and has no impacts. Otherwise, the GFM IBR operates in current-limiting mode. Following this principle, the power-angle relationship induced by the proposed VI-based CLC can be expressed as

$$\begin{aligned} P_{\text{VI}}^+ &= \text{Re}\left\{\dot{\boldsymbol{V}}_t^+ \,\text{conj}\left(\dot{\boldsymbol{I}}_f^+\right)\right\} \\ &\begin{cases} \dot{\boldsymbol{I}}_f^+ = \frac{V_{\text{ref}}e^{j\delta_p^+} - V_g^+ e^{j0}}{R_v^+ + R_g^+ + j\left(X_v^+ + X_g^+\right)} \\ \dot{\boldsymbol{V}}_t^+ = \dot{\boldsymbol{I}}_f^+\left(R_g^+ + jX_g^+\right) + V_g^+ e^{j0} \end{cases} \end{aligned} \tag{35}$$

It can be seen from (35) that the GFM IBR under the proposed VI-based CLC behaves as a voltage source in series with a virtual impedance in the power circuit representation.

2) Power-Angle Characterisation for CRS-Based Scheme: The current source representation is a common approach to characterise the power-angle profile of the GFM IBR using the angle-priority CRS-based CLC [41]. When the CRS-based CLC is activated, the active power output depends solely on the phase angle ϕ_{Ip}^+ under a specified V_g^+ and δ_p^+. The power-angle relationship can then be formulated as

$$P_{\text{CRS}}^+ = V_g^+ I_{\text{lim}}^+ \cos\left(\phi_{Ip}^+ - \delta_p^+\right) + \left(I_{\text{lim}}^+ R_g^+\right)^2 \tag{36}$$

To determine ϕ_{Ip}^+, equation (11) is rewritten in the phasor domain as

$$V_g^+ \sin\left(-\phi_{Ip}^+ + \phi_{z_g}^+ + \delta_p^+\right) = V_{\text{ref}} \sin\left(-\phi_{Ip}^+ + \phi_{z_g}^+\right) \tag{37}$$

Applying the same derivation steps from (11) to (13) yields:

$$\phi_{Ip}^+ = \phi_{z_g}^+ - \arctan\frac{-V_g^+ \sin\left(\delta_p^+\right)}{V_{\text{ref}} - V_g^+ \cos\left(\delta_p^+\right)} \tag{38}$$

It is important to note that the derived ϕ_{Ip}^+ in (38) is intended merely for transient stability analysis. In practical implementations of the proposed CRS-based CLC, the phase angle of the current reference $\boldsymbol{I}_{fdq}^{\text{ref}+}$ should follow the formulation in (17), rather than (38).

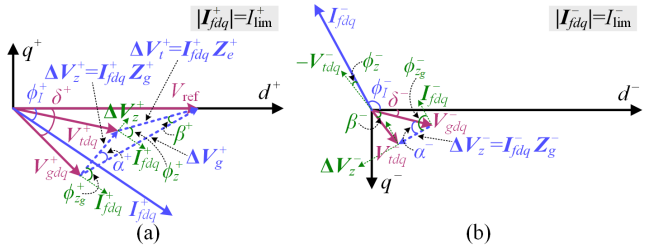


Fig. 9. Complex space vector diagram under asymmetric grid disturbances when $\phi_z^{+(-)} \neq \phi_{z_g}^{+(-)}$. (a) Positive-sequence. (b) Negative-sequence.

3) Equivalence of Power-Angle Characteristics for Different Methods: Fig. 8 shows both simulation and calculation power-angle curves under varying grid voltage magnitudes. The simulation results are obtained by setting the active power setpoints sufficiently larger than the maximum achievable active power output of the GFM IBR. The simulation adopts the default parameter settings specified in Section V. The calculation power-angle curves are derived from the theoretical models in (35) and (36).

Across all cases, the calculation results based on both CRS- and VI-based methods closely match the corresponding simulation results, validating the theoretical derivations from (34) to (38). The theoretical power-angle curves derived from the CRS-based method are identical to those of the VI-based counterpart, and their simulated profiles show close agreement, with only minor deviations due to inherent dynamics in time-domain simulations. This consistency confirms that, the voltage source representation of the proposed VI-based CLC is functionally equivalent to the current source representation of the proposed CRS-based CLC. Consequently, both methods deliver identical transient stability performance.

The observed equivalence can be interpreted via Fig. 7. As both schemes adhere to the CLC design guideline in Section III and fully utilise the inverter's current generation capacity, conditions $\angle \boldsymbol{Z}_e^+ = \angle \boldsymbol{Z}_g^+$ and $\left|\dot{\boldsymbol{I}}_f^+\right| = I_{\lim}^+$ consistently hold. Under a given V_g^+ and δ_p^+, the phasors $\dot{\boldsymbol{V}}_t^+$ and $\dot{\boldsymbol{I}}_f^+$ are therefore uniquely defined, independent of the CLC strategy, resulting in a unified power-angle characteristic.

D. Robustness Evaluation

One potential limitation of the proposed CLC method is its dependence on prior knowledge of the grid impedance angle $\phi_{z_g}^{+(-)}$ at the fundamental frequency. In practice, exact grid impedance parameters can be difficult to determine and may vary with different operating conditions. Therefore, the following analysis evaluates the performance of the proposed CLC scheme under errors in the obtained $\phi_{z_g}^{+(-)}$.

Assume that the phase angle actually used in the proposed CLC scheme is given by $\phi_z^{+(-)} = \arctan\left(k_e X_g^{+(-)}/R_g^{+(-)}\right)$, rather than the ideal value $\phi_{z_g}^{+(-)} = \arctan\left(X_g^{+(-)}/R_g^{+(-)}\right)$, where k_e is the error coefficient ($k_e > 0$ and $k_e \neq 1$). Under this context, the original condition in (6) is no longer satisfied, leading to $\alpha^{+(-)} \neq 0$. The resulting vector relationships are illustrated in Fig. 9.

Since $\alpha^{+(-)} \neq 0$, $\Delta \boldsymbol{V}_t^+$ is no longer collinear with $\Delta \boldsymbol{V}_z^+$ in Fig. 9(a), and $\boldsymbol{V}_{tdq}^-$ is not collinear with $\Delta \boldsymbol{V}_z^-$ in Fig. 9(b). However, the principles in (4) and (5) remain valid, implying that a smaller $\alpha^{+(-)}$ is desirable to maximise/minimise the positive/negative-sequence terminal voltage of the GFM IBR. From Figs. 9(a) and (b), $\alpha^{+(-)}$ can be expressed as

$$\alpha^{+(-)} = \phi_{z_g}^{+(-)} - \phi_z^{+(-)} - \beta^{+(-)} \tag{39}$$

Neglecting the positive term $\beta^{+(-)}$ in (39) yields $\alpha^{+(-)} \approx \phi_{z_g}^{+(-)} - \phi_z^{+(-)}$. Substituting $\phi_{z_g}^{+(-)} = \arctan(X_g^{+(-)}/R_g^{+(-)})$ and $\phi_z^{+(-)} = \arctan\left(k_e X_g^{+(-)}/R_g^{+(-)}\right)$, followed by algebraic simplifications, results in

$$\cos\left(\phi_{z_g}^{+(-)} - \phi_z^{+(-)}\right) = \frac{1 + k_e\left(\frac{X_g^{+(-)}}{R_g^{+(-)}}\right)^2}{\sqrt{\left[1 + \left(\frac{X_g^{+(-)}}{R_g^{+(-)}}\right)^2\right]\left[1 + \left(k_e\frac{X_g^{+(-)}}{R_g^{+(-)}}\right)^2\right]}} \tag{40}$$

For a given $X_g^{+(-)}/R_g^{+(-)}$ ratio, (40) allows quantitative analysis of the deviation $\alpha^{+(-)}$ caused by the error factor k_e. Considering a significant 50% deviation between the actual $X_g^{+(-)}/R_g^{+(-)}$ ratio and the used $k_e X_g^{+(-)}/R_g^{+(-)}$ ratio in the proposed CLC scheme ($k_e = 0.5$ or $k_e = 1.5$), the following cases are computed using (40):

1) $X_g^{+(-)}/R_g^{+(-)} = 10$: $\cos\alpha^{+(-)} \approx 1$ for both $k_e = 0.5$ and $k_e = 1.5$.
2) $X_g^{+(-)}/R_g^{+(-)} = 1$: $\cos\alpha^{+(-)} \approx 0.95$ for $k_e = 0.5$, and $\cos\alpha^{+(-)} \approx 0.98$ for $k_e = 1.5$.

Note that the computed $\cos\alpha^{+(-)}$ values are underestimated due to the omission of the positive term $\beta^{+(-)}$ in (39). Consequently, the actual values of $\cos\alpha^{+(-)}$ in the cases above exceed 0.95, indicating that $\alpha^{+(-)}$ remains relatively small and close to zero. This analysis confirms that the proposed CLC method largely maintains the condition in (6), even when the adopted $X^{+(-)}/R^{+(-)}$ ratio in $\phi_z^{+(-)}$ deviates significantly from the actual $X_g^{+(-)}/R_g^{+(-)}$ ratio. Therefore, the proposed method is robust against parameter uncertainties.

V. Performance Evaluation

Simulations in Matlab/Simulink and CHIL experiments are conducted on a 10 kVA three-phase GFM IBR with the follwing default parameters: $R_f = 0.1\ \Omega$, $L_f = 5$ mH, $C_f = 15\ \mu$F, $m_p = 0.01$ p.u., $m_q = 0.1$ p.u., $P_0 = 0.7$ p.u., $Q_0 = 0$, $K^- = 3$, and $I_{\lim} = 1.5$ p.u., as recommended in [42]. A low APC droop coefficient (m_p) is adopted to mitigate the potential small-signal stability issues associated with large droop gains in GFM IBRs operating under the sequence-decomposed control framework, as discussed in [14], [15]. Since the study mainly focuses on CLC schemes implemented within the inner voltage and current control loops, variations in APC droop gain have negligible impact on CLC behaviour during disturbances, due to the bandwidth separation between the outer power control and inner control layers.

The depicted system in Fig. 1 is leveraged as the testbench circuit, with a step-up transformer (ratio: $0.38/10$ kV). A default 0.3 p.u. positive- (negative-) sequence grid reactance

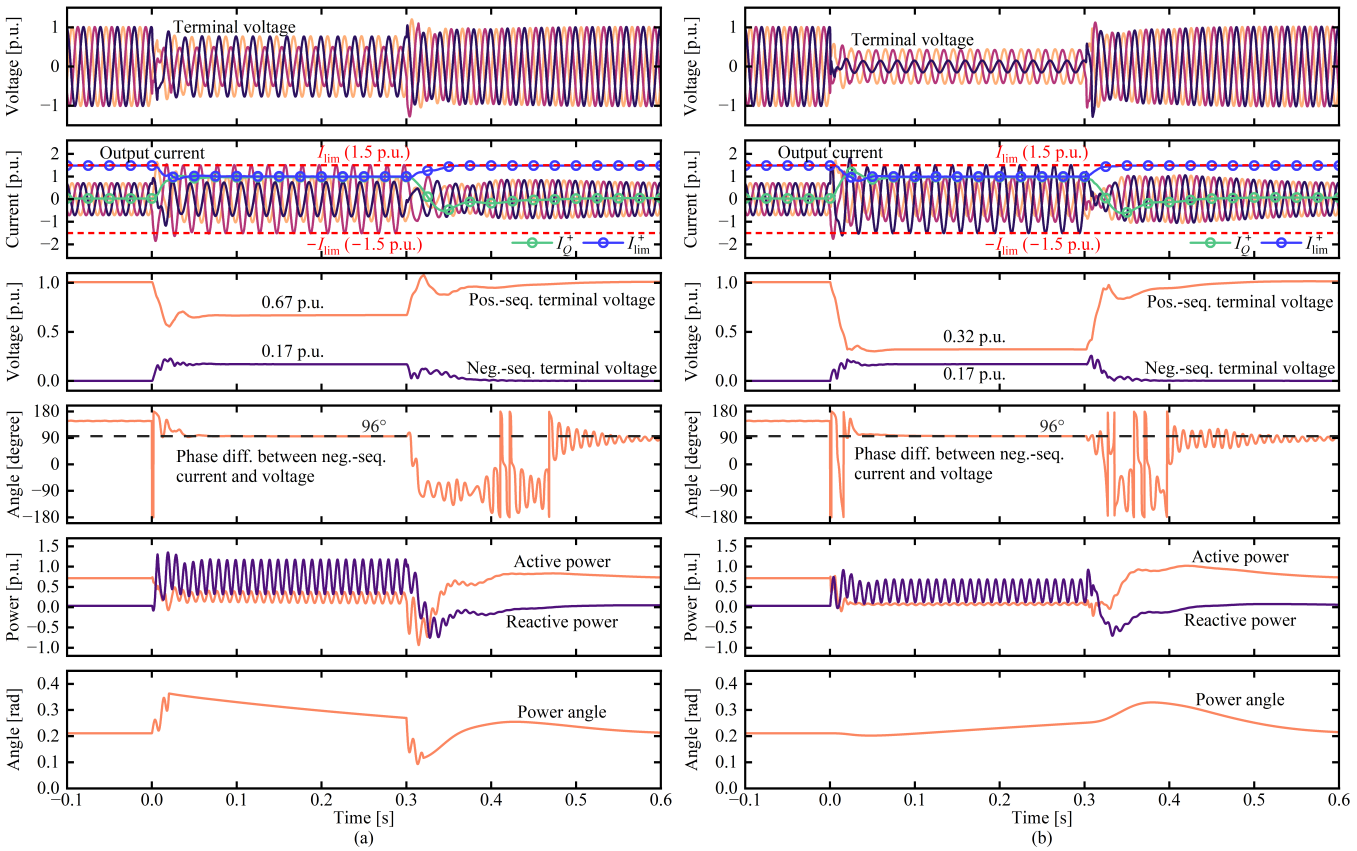


Fig. 10. Simulation results for asymmetric grid disturbances. (a) Two-phase voltage sag plus $-60°$ phase angle jump. (b) Two-phase-to-ground fault.

($X_g^{+(-)} = 0.3$ p.u.) is used in the grid impedance to emulate an AC grid with a short-circuit ratio (SCR) of 3.33, with $\mathbf{Z}_{g1} : \mathbf{Z}_{g2} = 1 : 2$, and the positive- (negative-) sequence grid resistance is determined by the $X_g^{+(-)}/R_g^{+(-)}$ ratio.

In all subsequent case studies, the proposed angle-priority CRS method in Section IV is used for illustration, and the considered grid disturbances always activate the CLC.

A. Dynamic Response Under Asymmetric Grid Disturbances

Asymmetric grid disturbances, including grid voltage sags, grid phase angle jumps, and short-circuit faults, are applied to the testbench circuit to evaluate the performance of the proposed CLC scheme. A typical transmission system is simulated by setting $X_g^{+(-)}/R_g^{+(-)} = 10$ ($\phi_{z_g}^{+(-)} \approx 84°$).

Fig. 10 presents the dynamic response of the GFM IBR under the proposed CRS-based CLC, configured with $\phi_z^{+(-)} = \phi_{z_g}^{+(-)}$. Specifically, Fig. 10(a) displays the response to a two-phase grid voltage sag combined with a $-60°$ phase angle jump, where the perturbed phase voltage dips to 0.1 p.u. Fig. 10(b) shows the response to a two-phase-to-ground fault, with the fault location indicated in Fig. 1. During each disturbance (lasts 300 ms), the active power setpoint is reduced to 0.15 p.u. to enhance postfault transient stability.

The terminal voltage profiles in Figs. 10(a) and (b) confirm the stable operation of the GFM IBR throughout the disturbance. At the onset of the disturbance, the proposed CLC scheme acts rapidly to suppress the output current to $I_{\rm lim}$ (1.5 p.u.). Although a brief overshoot occurs due to the controller's inherent response delay, it fully decays within half a cycle and remains well below the 2.0 p.u. threshold of the inverter overcurrent protection. Moreover, the proposed CLC exercises the IBR's full range of current generation capacity by regulating the phase current to $I_{\rm lim}$, ensuring compliance with GCs. By applying (20) and (26), $I_{\rm lim}^+$ and I_Q^+ are obtained, showing close alignment in the quasi-steady state in Fig. 10, consistent with the derivation in (30). This shows that the proposed CLC manipulates $\left|I_Q^+\right|$ to the positive-sequence current limit $I_{\rm lim}^+$, thereby meeting the GC requirements for positive-sequence reactive current injection.

The positive- and negative-sequence terminal voltage magnitudes are also plotted in Figs. 10(a) and (b). The results indicate that the GFM IBR acts as a sinker to counter unbalance in its terminal voltage, resulting in improved positive-sequence voltage magnitudes and suppressed negative-sequence voltage magnitudes, even under severe asymmetric fault conditions.

The fourth subplots in Figs. 10(a) and (b) depict the phase angle difference between the negative-sequence output current and terminal voltage. The results show that the negative-sequence output current leads the negative-sequence terminal voltage by about 96°, remaining within the GC-specified range of 90° to 100°. The slight deviation of 6° from 90° is caused by the setting $\phi_z^- = 84°$ in (18). Oscillations are observed after 0.3 s. This occurs as the system transitions to a symmetric operating condition following fault clearance, where all negative-sequence components vanish. Under such conditions, both the negative-sequence current and terminal voltage approach zero magnitude. Since the phase angle of a zero vector is undefined, the resulting angle difference

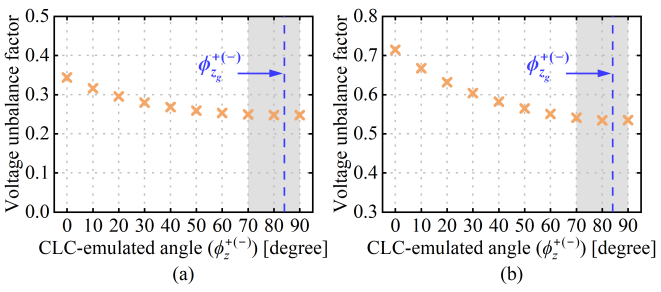


Fig. 11. Voltage unbalance factor under different $\phi_z^{+(-)}$. (a) Two-phase voltage sag plus $-60°$ phase angle jump. (b) Two-phase-to-ground fault.

becomes mathematically indeterminate, which explains the observed fluctuations.

As also shown in Fig. 10, the reactive current increases naturally at the onset of the disturbance, indicating that the proposed CLC preserves the inherent voltage source characteristics of the GFM IBR. The power angle δ_p^+, defined as the phase difference between the positive-sequence internal voltage phasor and the terminal voltage phasor, is presented in the last subplot of Figs. 10(a) and (b). In Fig. 10(a), the power angle exhibits a sharp increase following the applied backward phase-angle jump, then gradually decreases prior to fault clearance. In contrast, Fig. 10(b) shows a gradual increase in δ_p^+ during the fault, caused by the slight active power imbalance under this severe grid fault. Across both cases, no divergent behaviour is observed in the power angle trajectory, and the system consistently returns to its pre-fault operating state without instability.

Fig. 11 presents the voltage unbalance factor (VUF) results, defined as the ratio of negative- to positive-sequence voltage magnitudes [35], under different $\phi_z^{+(-)}$ settings. The disturbance conditions are identical to those in Fig. 10. As $\phi_z^{+(-)}$ increases in both Figs. 11(a) and (b), the VUF decreases, reflecting improved positive-sequence terminal voltage and suppression of negative-sequence terminal voltage. Optimal performance is achieved when $\phi_z^{+(-)}$ lies within 80°–90°, aligning with the AC line impedance angle $\phi_{z_g}^{+(-)} = 84°$ used in Fig. 11. As noted in Section IV.A, a $\phi_z^{+(-)}$ within 80°–90° implies that the equivalent impedance emulated by the CLC has a phase angle within the same range. Thus, the observed optimal $\phi_z^{+(-)}$ range in Fig. 11 validates the theoretical analysis in Section III, confirming that aligning $\angle \boldsymbol{Z}_e^{+(-)}$ with $\angle \boldsymbol{Z}_g^{+(-)}$ enhances GFM capability. Furthermore, VUF remains nearly constant for $\phi_z^{+(-)}$ between 70° and 90°, as shown in Fig. 11. This indicates that, even with a suboptimal $\phi_z^{+(-)}$ deviating from $\phi_{z_g}^{+(-)}$, the proposed CLC scheme largely preserves the enhanced GFM capability, thereby reinforcing the robustness analysis in Section IV.D.

B. Comparative Transient Stability Analysis

For transient stability analysis, the default settings are considered, with $X_g^+/R_g^+ = 10$.

1) Comparison with the Existing CRS- and VI-based CLC Methods: Fig. 12 shows theoretical phase-angle curves of the GFM IBR under different CLC strategies. As a benchmark, the case without any CLC is also included. In addition to

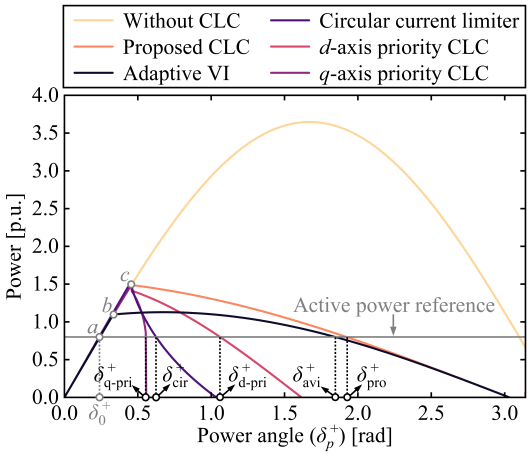


Fig. 12. Power-angle curves under different CLC strategies.

the power-angle result derived from the proposed method (see Section IV.C), four representative existing approaches are compared, including the d-axis priority CRS-based method [38], the q-axis priority CRS-based method [11], the circular current limiter [34], and the optimised adaptive VI with fixed proportional gains [28] (the proportional gain is tuned to avoid continuous overcurrent under a three-phase bolted fault at the IBR terminal). For brevity, the only difference among these methods lies in the specific CLC implementation, while their basic power, voltage, and current control loops remain the same (the performance of each method may be further adjusted by tuning the corresponding control parameters). Under this condition, and for a given active power reference $P_0 = 0.8$ p.u.), all methods yield the same initial phase angle δ_0^+ and the same operating point 'a' in Fig. 12.

The transition from unsaturated to saturated current condition occurs at point 'b' for the VI-based method and at point 'c' for CRS-based methods. This difference arises because the CRS-based CLC activates when $\left|\boldsymbol{I}_{fdq}^+\right| > I_{\text{lim}}^+$, while the VI-based method activates earlier, at $\left|\boldsymbol{I}_{fdq}^+\right| > I_{\text{th}}^+$. Consequently, the peak value of the power-angle curve for CRS-based methods is higher than that of the VI-based method.

Due to the distinct CLC mechanisms, the intersection points between the active power reference line and the power-angle curves vary across methods. The associated phase angles are denoted as $\delta_{\text{q-pri}}^+$, δ_{cir}^+, $\delta_{\text{d-pri}}^+$, δ_{avi}^+, and δ_{pro}^+ in Fig. 12. Considering a three-phase bolted short-circuit fault occurs at the IBR terminal, the active power output of the GFM IBR drops to zero. The corresponding critical clearing time (CCT) can be approximated using the method reported in [43]:

$$t_x^{\text{CCT}} = \frac{\delta_x^+ - \delta_0^+}{m_p \omega_0 P_0} \tag{41}$$

where $x \in \{\text{pro, d-pri, q-pri, cir, avi}\}$.

It can be seen from (41) that the calculated CCT has a linear relationship with δ_x^+. Therefore, the δ_{pro}^+ yielded by the proposed method theoretically corresponds to the largest CCT, indicating the best transient stability margin among the methods compared. This superior performance is attributed to two core design considerations involved in the proposed CLC method: i) matching the phase angle of the equivalent impedance emulated by the CLC with that of the

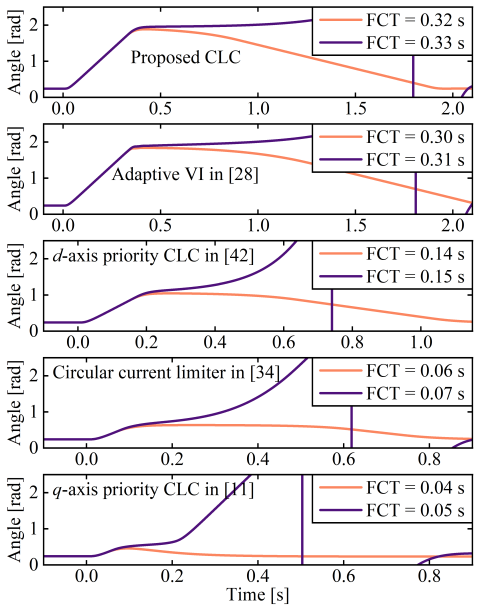


Fig. 13. Simulation results for power angle under different CLC strategies.

grid impedance; ii) maximising the IBR's current generation capacity during current limitation. The first consideration enhances voltage support during disturbances, while the second consideration enables greater active power output by fully utilising the inverter's current limit. Together, they improve the power-angle profile and significantly enhance transient stability. Among the referenced methods, under symmetric fault conditions, [11], [34], and [38] incorporate only the second consideration, while [28] applies merely the first. As a result, these methods underperform the proposed method in terms of transient stability.

To verify the theoretical analysis, Fig. 13 shows the simulation power angle trajectories of the GFM IBR under different CLC strategies. The three-phase bolted short-circuit fault is applied at the IBR terminal and the active power setpoint is kept 0.8 p.u. throughout the period, with $m_p = 0.02$ p.u.

Fig. 13 includes cases where the fault clearing time (FCT) is slightly shorter and slightly longer than the corresponding CCTs. It can be observed that the proposed CLC method obtains the maximum CCT (between 0.32 s and 0.33 s), confirming its superior transient stability relative to the methods in [11], [28], [34], and [38]. These results align with the theoretical findings in Fig. 12 and underscore the importance of simultaneously involving the above two considerations in the CLC design for enhanced transient stability.

Finally, it is worth noting that the CLC strategies in [11], [28], [34], and [38] violate modern GC requirements, such as those outlined in [21] and [22], by generating non-compliant negative-sequence current angles. This limitation arises from their adoption of sequence-coupled control frameworks. Additionally, the fixed-gain VI-based approach in [28] may underutilise the inverter's current generation capacity under certain fault conditions, resulting in further GC non-compliance.

2) Comparison with the Emerging Cross-Forming Control Method: A recently proposed current-limiting concept, termed cross-forming (CFM) control, has been introduced in [37]. Unlike conventional VI- or CRS-based CLC strategies, the

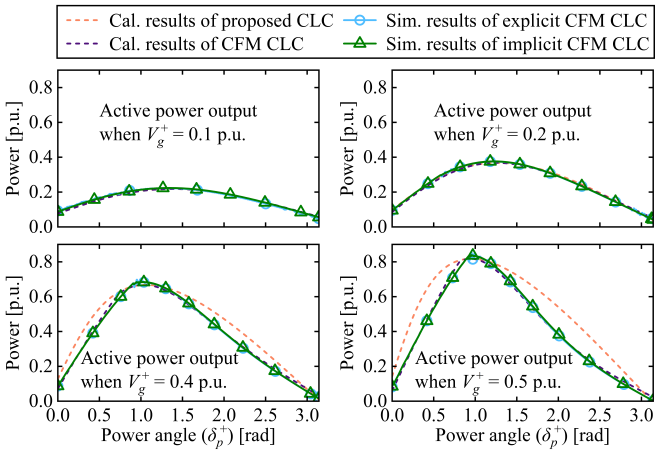


Fig. 14. Power-angle curves of the proposed and the CFM methods.

CFM method aims to independently regulate the voltage phase and current magnitude during disturbances, while allowing the voltage magnitude and current phase to be dictated by the external power grid.

As documented in [37], the CFM control can be implemented in both explicit and implicit forms. Its current-limiting mechanism is to adjust the input signals of the voltage control loop—either the internal voltage reference or the voltage feedback—based on the output current state. Following this principle, both the explicit and implicit CFM implementations can be equivalently interpreted as reducing the internal voltage magnitude from V_{ref} to $k_c V_{\text{ref}}$, where k_c is an equivalent scaling factor ranging from 0 to 1. The purpose of this voltage reduction is to ensure the output current is constrained within I_{lim} during disturbances. Thus, the following relationship holds when the CFM method is activated:

$$\left| \frac{k_c V_{\text{ref}} e^{j\delta_p^+} - V_g^+ e^{j0}}{R_v^+ + R_g^+ + j\left(X_v^+ + X_g^+\right)} \right| = I_{\text{lim}}^+ \tag{42}$$

Here, R_v^+ and X_v^+ denote the virtual resistance and reactance used in the VAC within the CFM framework, typically fixed values corresponding to the static virtual admittance [37]. Given a specified grid voltage magnitude V_g^+ and power angle δ_p^+ that activate the CFM control, k_c can be computed using (42). The corresponding power-angle relationship of the GFM IBR under the CFM operation is then expressed as

$$\begin{aligned} P_{\text{CFM}}^+ &= \text{Re}\left\{ \dot{\boldsymbol{V}}_t^+ \,\text{conj}\left(\dot{\boldsymbol{I}}_f^+ \right) \right\} \\ &\begin{cases} \dot{\boldsymbol{I}}_f^+ = \frac{k_c V_{\text{ref}} e^{j\delta_p^+} - V_g^+ e^{j0}}{R_v^+ + R_g^+ + j\left(X_v^+ + X_g^+\right)} \\ \dot{\boldsymbol{V}}_t^+ = \dot{\boldsymbol{I}}_f^+ \left(R_g^+ + jX_g^+\right) + V_g^+ e^{j0} \end{cases} \end{aligned} \tag{43}$$

Fig. 14 compares the power-angle curves derived from both the proposed method and the CFM approach. Simulation results for both explicit and implicit CFM implementations are included. Both methods use the actual active power rather than the virtual one used in [37], [44]. Since $0 < k_c < 1$, Fig. 14 mainly shows results under grid voltage sag conditions.

As shown in Fig. 14, the simulation results for both the explicit and implicit CFM methods align well with the derived power-angle curves calculated using (42) and (43), validating the correctness of the theoretical analysis. Comparisons of

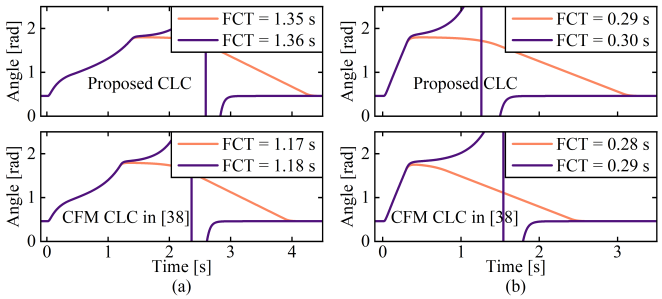


Fig. 15. Simulation results for power angle under the proposed and the CFM methods. (a) $V_g^+ = 0.5$ p.u. (b) $V_g^+ = 0.1$ p.u.

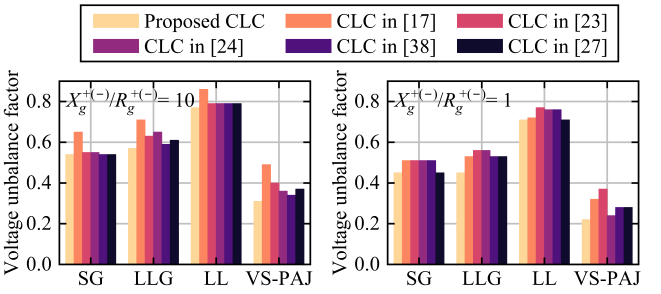


Fig. 16. Voltage unbalance factor results yielded by different CLC strategies under diverse asymmetric disturbances.

the power-angle results between the proposed method and the CFM method provide three main observations: i) for different V_g^+, the peak values of the power-angle curves are identical for both methods, ii) under mild voltage sags (higher V_g^+), the proposed method obtains higher active power output across most δ_p^+, except at the peak, iii) under severe sags (lower V_g^+), both methods exhibit similar power-angle behaviour. The first observation is expected, as both methods fully utilise the inverter's current generation capacity. The second and third findings, however, result from the distinct current-limiting mechanisms employed by the two approaches. Specifically, the proposed method limits the output current by adjusting the CLC-emulated impedance, whereas the CFM method achieves current limitation by equivalently modifying the internal voltage reference.

To validate the above analysis, simulations are performed under grid voltage sag conditions. Fig. 15 shows the power-angle trajectories for the proposed method and the CFM method (explicit implementation), both configured with the same voltage control (VAC), and identical power and current control. Two grid conditions are evaluated: $V_g^+ = 0.5$ p.u. and $V_g^+ = 0.1$ p.u., with the active power setpoint fixed at 0.9 p.u. throughout the test ($m_p = 0.02$ p.u.). Each case includes scenarios where the FCT is shorter and longer than the respective CCTs. Under moderate voltage sag [see Fig. 15(a)], the CCT of the proposed method (between 1.35 s to 1.36 s) exceeds that of the CFM method. In contrast, under severe sag conditions [see Fig. 15(b)], the difference in CCTs narrows to about 10 ms. These results confirm the theoretical analysis and demonstrate the improved transient stability of the proposed method.

In terms of GC compliance, the CFM method employs control formulations based on phase currents, as discussed in [37]. This design enables the GFM IBR to fully utilise its current generation capability under both symmetric and asymmetric disturbances. In the CFM framework in [37], a purely inductive virtual admittance is assumed in the adopted VAC. As a result, condition (24) also holds for the CFM method, provided that V_{ref} is replaced by $k_c V_{\text{ref}}$. Following the derivations from (25) to (29), condition (30) is similarly obtained for the CFM control, demonstrating compliance with the positive-sequence reactive current injection requirements specified in [21], [22]. Additionally, the CFM method incorporates the negative-sequence voltage mitigation mode developed in accordance with IEEE Std. 2800-2022. Consequently, the CFM approach also satisfies GC requirements related to negative-sequence current injection.

C. Comparative GFM Capability Analysis

As specified in [5]–[9], under asymmetric grid disturbances, GFM IBRs are expected to maintain balanced terminal voltages as much as possible to enhance GFM capability. Accordingly, the VUF is used to evaluate the GFM capability of the GFM IBR under different CLC strategies.

The VUF results are analysed under two different $X_g^{+(-)}/R_g^{+(-)}$ ratios: 10 and 1. The former represents a highly inductive AC network, while the latter corresponds to a system with comparable grid resistance and reactance. Four types of asymmetric disturbances are applied to the testbench circuit in Fig. 1: single-phase-to-ground fault (SLG), two-phase-to-ground fault (LLG), phase-to-phase fault (LL), and two-phase grid voltage sag with -60° phase angle jump (VS-PAJ).

The CLC strategies in [11], [28], [34], and [38] are based on the sequence-coupled control framework and are primarily designed for symmetric grid conditions. These methods are therefore excluded from the following analysis.

All CLC schemes included in the comparison are originally developed within the sequence-decomposed control framework. For consistency, all schemes are implemented in the dq-frame and use identical APC, RPC, and VCC. In addition, the current limit I_{lim} is uniformly set to 1.2 p.u. to ensure all CLC schemes are activated during disturbances. The considered CLC schemes are categorised into three groups:

- *Effective current-limiting methods that fail to satisfy GC requirements [17]–[19].* A typical approach configures an elliptical limiter to saturate current references during current limitation, as proposed in [17].
- *GC-compliant methods that do not target enhancement of GFM capability [23]–[26], [37].* Representative schemes include the integration of VI with CRS [23], VA with CRS [24], and the CFM method [37].
- *Enhanced GFM capability scheme but lacking formal mathematical validation and full GC compliance [27].* The adaptive VI method in [27], originally formulated in the $\alpha\beta$-frame, is adapted here to the dq-frame.

Fig. 16 shows the VUF results under various CLC strategies. Under both high X/R ratio and low X/R ratio conditions, the proposed CLC method (represented by the first bar in each subplot) always achieves the lowest VUF, indicating the most balanced terminal voltage under asymmetric disturbances. This outcome reflects the enhanced GFM capability achieved by the proposed method compared to existing alternatives.

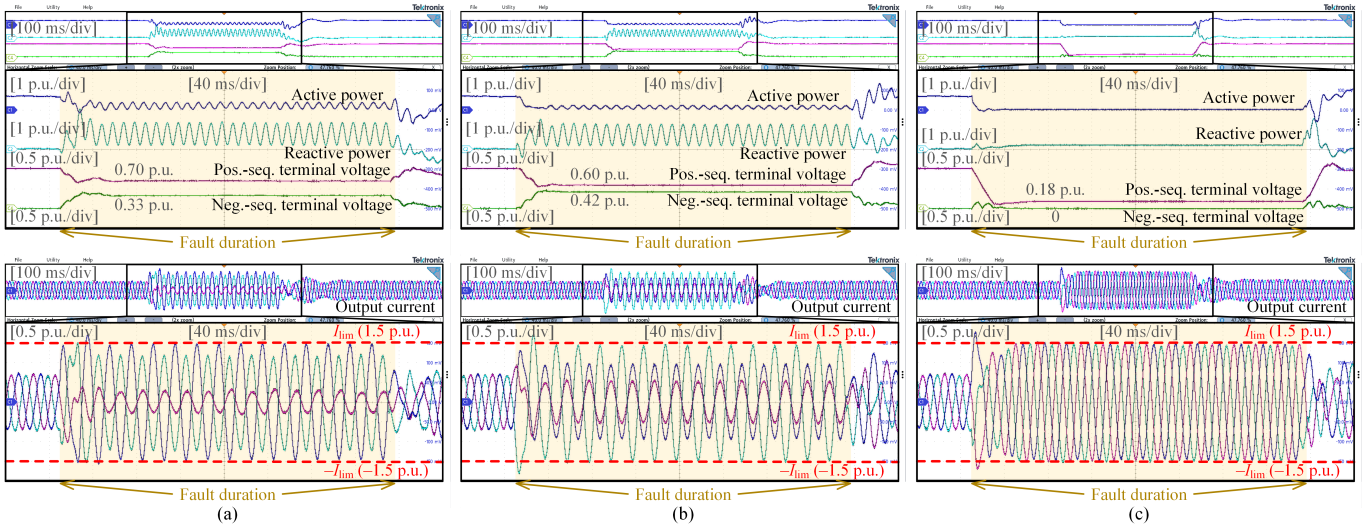


Fig. 17. Experimental results when SCR = 3.33 (Indicators C1, C2, C3, and C4 on the far left of each graph represent the datum points of the corresponding coloured curves; div refers to the measurement scale per grid division on the oscilloscope, the same for Figs. 18 and 19). (a) Single-phase-to-ground fault. (b) Phase-to-phase fault. (c) Three-phase-to-ground fault.

In the highly inductive network case ($X_g^{+(-)}/R_g^{+(-)} = 10$), the proposed CLC method satisfies the GC requirements in terms of i) full utilisation of the IBR's current generation capacity, and ii) proper injection of positive- and negative-sequence reactive currents. Furthermore, the method incorporates the proposed CLC design principle of aligning the phase angle of the CLC-emulated impedance with that of the grid impedance. By contrast, the method in [17] satisfies neither the current injection criteria defined by GCs nor the proposed CLC design guideline. Consequently, it consistently exhibits the highest VUF results in the first subplot. The approaches in [23], [24], [37] fulfil GC requirements related to current utilisation and reactive current injection, yet do not apply the proposed design principle. Meanwhile, the method in [27] adopts the empirical design strategy that conceptually aligns with the proposed principle, yet fails to fully utilise the inverter's current generation capacity. These four approaches yield comparable VUF results, though all underperform relative to the proposed method.

Under low X/R ratio conditions ($X_g^{+(-)}/R_g^{+(-)} = 1$), the strict reactive current injection requirements specified in [21], [22]—which target transmission-level systems—are no longer appropriate. Instead, both active and reactive current injections must be considered, with their ratio determined by the network's X/R value [40]. The proposed CLC method accommodates this need by applying the proposed CLC design guideline. Therefore, it continues to deliver the lowest VUF among all compared strategies. The method in [17] utilises an elliptical limiter that reduces current reference magnitudes without altering their phase angles. During current limitation, this results in a strong resistive behaviour in both the positive- and negative-sequence domains, similar to the circular limiter discussed in [34]. Consequently, the VUF achieved by this method is lower than those of [23], [24], [37] in most cases, which focus solely on GC compliance within transmission networks. Meanwhile, the method in [27] produces slightly higher VUF values than the proposed method due to its conservative current utilisation.

D. Controller Hardware-in-the-Loop Validation

A CHIL test platform, comprising an OPAL-RT 5600 real-time digital simulator (for power circuit emulation) and a TMS320F28379D-based DSP controller (for control algorithm implementation), is established to further corroborate the proposed CLC method. During testing, the OPAL-RT 5600 outputs analog signals, including terminal voltage, output current, and grid current, to the DSP controller, which in turn generates PWM digital signals for the OPAL-RT 5600. This configuration enables a relatively realistic evaluation of the proposed control strategy.

Three SCR levels are tested: 1.43 (ultra-weak grid), 3.33 (moderate grid), and 8 (strong grid), with $X_g^{+(-)}/R_g^{+(-)} = 10$. During disturbances, the active power setpoint is reduced to 0.15 p.u. to mitigate active power imbalance and improve post-fault transient stability. All other parameters follow the default settings specified at the beginning of Section V. For all three SCR conditions, the positive- and negative-sequence terminal voltage magnitudes are obtained using the symmetrical component transformation, and their linear sum is determined by the fault type, network impedance, and operating condition.

Fig. 17 shows experimental results for three fault types, each lasting 300 ms. The tests are conducted under the moderate SCR condition of 3.33, consistent with the simulation setup. Under the proposed CLC method, the active and reactive power respond instantaneously in all fault scenarios. Additionally, improved positive-sequence voltage profiles and suppressed negative-sequence components are observed, indicating that the terminal voltage remains as balanced as possible during asymmetric faults. Although brief overshoots occur in the output current, they attenuate rapidly and constantly stay below the threshold of the inverter overcurrent protection (2.0 p.u.). Meanwhile, the quasi-steady-state current remains within the imposed current limit and is maximised to fully utilise the inverter's current generation capacity (1.5 p.u.). After fault clearances, the GFM IBR exits the current-limiting mode and promptly returns to the normal operation, demonstrating

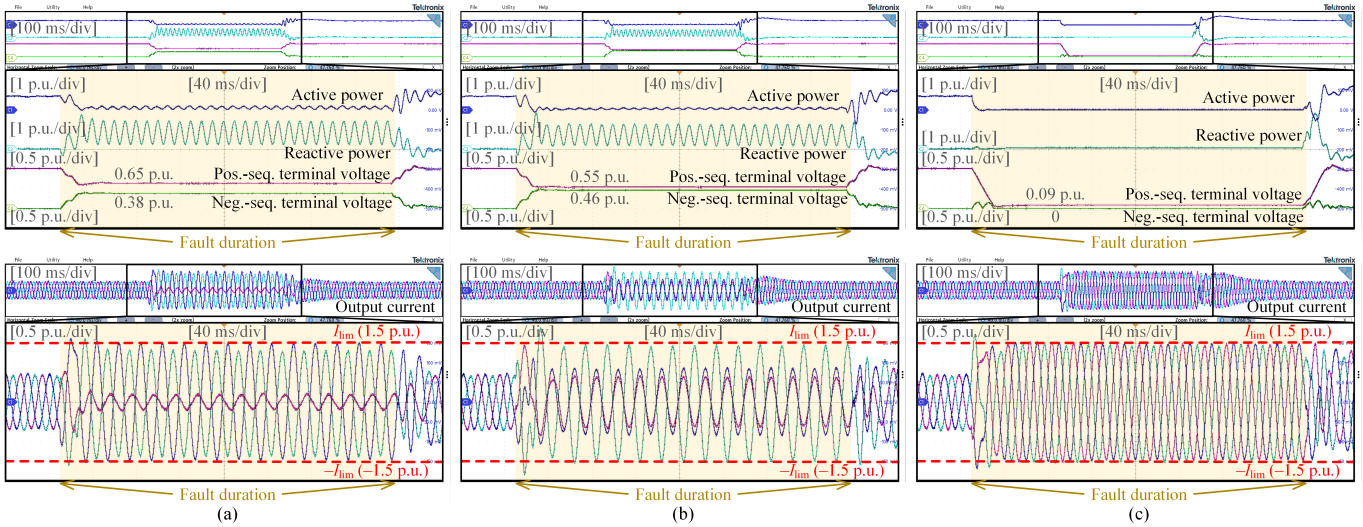


Fig. 18. Experimental results when SCR = 8. (a) Single-phase-to-ground fault. (b) Phase-to-phase fault. (c) Three-phase-to-ground fault.

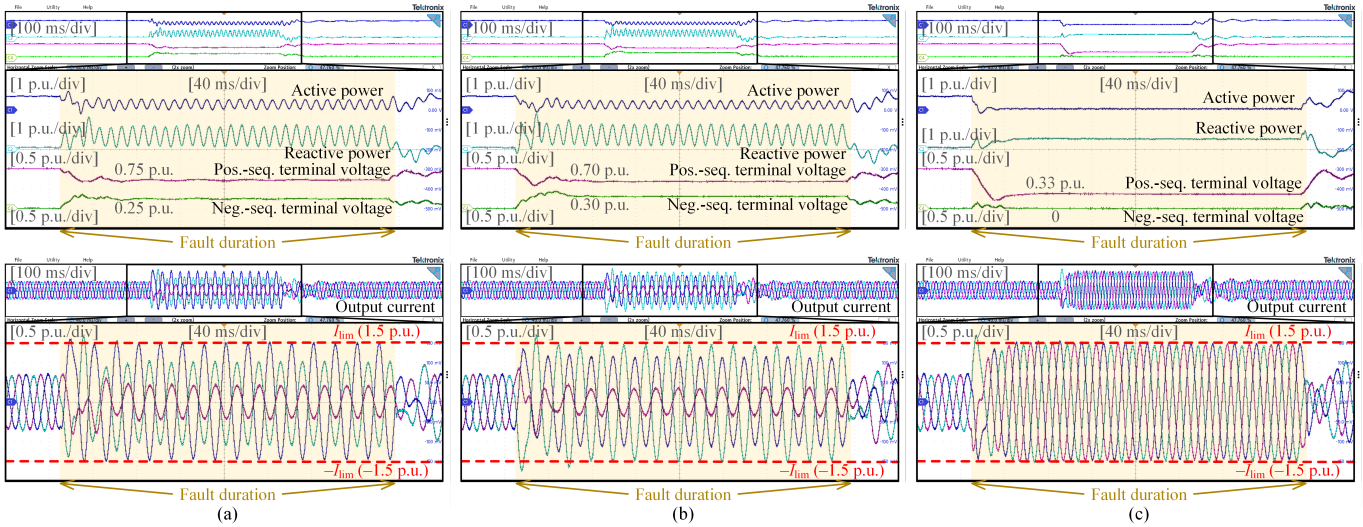


Fig. 19. Experimental results when SCR = 1.43. (a) Single-phase-to-ground fault. (b) Phase-to-phase fault. (c) Three-phase-to-ground fault.

favourable post-fault recovery performance.

Compared to the moderate grid case, experimental results under strong grid conditions in Fig. 18 reveal a slight reduction in the positive-sequence terminal voltage and a corresponding increase in the negative-sequence voltage under the same fault type. This behaviour can be interpreted using the vector diagram in Fig. 5, where a stronger grid results in a smaller radius of $I_{\text{lim}}^{+(-)} \left| \boldsymbol{Z}_g^{+(-)} \right|$. Consequently, the magnitude of the positive-sequence terminal voltage decreases, while that of the negative-sequence component increases. Across all three fault types in the strong grid, the GFM IBR constrains the output current to I_{lim} and achieves synchronisation with no sustained oscillations following fault clearance.

Finally, Fig. 19 shows the test results under an ultra-weak grid condition (SCR = 1.43). Prior to the fault, sufficient reactive power is supplied to maintain the positive-sequence terminal voltage at 1.0 p.u., enabling active power transfer at the predefined level P_0. During the fault, the active power output decreases significantly. In contrast, the GFM IBR provides a strong reactive power injection to support terminal voltage. Due to the high grid impedance in ultra-weak grids, the highest positive-sequence voltage and the lowest negative-sequence voltage among the three grid cases are observed under identical fault scenarios (see Figs. 17, 18 and 19). After fault clearance, the GFM IBR returns to its pre-fault operating point without excessive oscillations or instability.

VI. Conclusion

Per recent technical specifications, GFM IBRs are required to maintain terminal voltages as balanced as possible during asymmetric disturbances to enhance GFM capability. Complex space vector analysis reveals that the requirement can be achieved by aligning the phase angle of the CLC-emulated impedance with that of the AC grid impedance. This principle is applied to the VI-based CLC via direct configuration of the control X/R ratio, and extended to the CRS-based CLC by adaptively adjusting the current reference angle based on dq-axis terminal voltage dynamics. Rigorous theoretical analysis demonstrates that both CLC schemes comply with modern GCs, exhibit identical transient stability, and remain robust under parameter uncertainties. Comparative simulation studies against existing CLC strategies, along with CHIL experiments, validate the effectiveness of the proposed method in achieving enhanced transient stability and GFM capability. Future work

will focus on improving the performance of the proposed CLC schemes in complex power systems with multiple IBRs to enable coordinated global voltage support.

REFERENCES

[1] J. Matevosyan *et al.*, "Grid-forming technology in energy systems integration," Energy Systems Integration Group (ESIG), Tech. Rep., Mar. 2022.

[2] "Great Britain grid forming best practice guide," National Energy System Operator (NESO), Tech. Rep., Apr. 2023.

[3] H. Wu and X. Wang, "Control of grid-forming VSCs: A perspective of adaptive fast/slow internal voltage source," *IEEE Trans. Power Electron.*, vol. 38, no. 8, pp. 10 151–10 169, Aug. 2023.

[4] A. Ordono, A. Sanchez-Ruiz, M. Zubiaga, F. J. Asensio, and J. A. Cortajarena, "Current limiting strategies for grid forming inverters under low voltage ride through," *Renew. Sustain. Energy Rev.*, vol. 202, p. 114657, Sep. 2024.

[5] "Voluntary specification for grid-forming inverters," Australian Energy Market Operator (AEMO), Tech. Rep., May 2023.

[6] "MISO grid-forming battery energy storage capabilities, performance, and simulation test requirements proposal," Midcontinent Independent System Operator (MISO), Tech. Rep., Jul. 2024.

[7] "Specific study requirements for grid energy storage systems," FINGRID, Tech. Rep., Jun. 2023.

[8] "Requisitos técnicos mínimos para recursos basados en inversores grid-forming," Coordinador Eléctrico Nacional (CEN), Tech. Rep., Oct. 2024.

[9] "UNIFI specifications for grid-forming inverter-based resources–Version 2," UNIFI Consortium, UNIFI-2024-2-1, Tech. Rep., Mar. 2024.

[10] N. Baeckeland, D. Chatterjee, M. Lu, B. Johnson, and G.-S. Seo, "Overcurrent limiting in grid-forming inverters: A comprehensive review and discussion," *IEEE Trans. Power Electron.*, vol. 39, no. 11, pp. 14 493–14 517, Nov. 2024.

[11] S. P. Me *et al.*, "Transient stability analysis of virtual synchronous generator equipped with quadrature-prioritized current limiter," *IEEE Trans. Power Electron.*, vol. 38, no. 9, pp. 10 547–10 553, Sep. 2023.

[12] B. Fan, T. Liu, F. Zhao, H. Wu, and X. Wang, "A review of current-limiting control of grid-forming inverters under symmetrical disturbances," *IEEE Open J. Power Electron.*, vol. 3, pp. 955–969, 2022.

[13] A. Azizi and A. Hooshyar, "Fault current limiting and grid code compliance for grid-forming inverters—Part I: Problem statement," *IEEE Trans. Sustain. Energy*, vol. 15, no. 4, pp. 2486–2499, Oct. 2024.

[14] Q. Liu, Y. Liang, Z. Zhang, M. Wang, and X. Zhao, "Small-signal stability of sequence-decomposed grid-forming IBRs with DC-link voltage dynamics," *IEEE Trans. Power Electron.*, vol. 40, no. 2, pp. 2971–2986, Feb. 2025.

[15] Q. Liu, Y. Liang, Z. Zhang, M. Wang, and X. Zhao, "Small-signal synchronization stability of sequence-decomposed grid-forming IBRs," *IEEE Trans. Ind. Electron.*, vol. 72, no. 6, pp. 6049–6060, Jun. 2025.

[16] B. Mahamedi, M. Eskandari, J. E. Fletcher, and J. Zhu, "Sequence-based control strategy with current limiting for the fault ride-through of inverter-interfaced distributed generators," *IEEE Trans. Sustain. Energy*, vol. 11, no. 1, pp. 165–174, Dec. 2020.

[17] J. Freytes, J. Li, G. de Préville, and M. Thouvenin, "Grid-forming control with current limitation for MMC under unbalanced fault ride-through," *IEEE Trans. Power Del.*, vol. 36, no. 3, pp. 1914–1916, Jun. 2021.

[18] Z. Li, K. W. Chan, J. Hu, and S. W. Or, "An adaptive fault ride-through scheme for grid-forming inverters under asymmetrical grid faults," *IEEE Trans. Ind. Electron.*, vol. 69, no. 12, pp. 12 912–12 923, Dec. 2022.

[19] H. Zhang, R. Liu, C. Xue, and Y. Li, "Active power enhancement control strategy of grid-forming inverters under asymmetrical grid faults," *IEEE Trans. Power Electron.*, vol. 39, no. 1, pp. 1447–1459, Jan. 2024.

[20] "The serviced grid code – Issue 6 revision 25," *National Energy System Operator (NESO)*, Jul. 2024.

[21] "Technical requirements for the connection and operation of customer installations to the high voltage network (TAR high voltage)," *VDE-AR-N 4120*, Nov. 2018.

[22] "IEEE Standard for interconnection and interoperability of inverter-based resources (IBRs) interconnecting with associated transmission electric power systems," *IEEE Std. 2800-2022*, Apr. 2022.

[23] M.-A. Nasr and A. Hooshyar, "Controlling grid-forming inverters to meet the negative-sequence current requirements of the IEEE Standard 2800-2022," *IEEE Trans. Power Del.*, vol. 38, no. 4, pp. 2541–2555, Aug. 2023.

[24] Y. Zhang, C. Zhang, M. Molinas, and X. Cai, "Control of virtual synchronous generator with improved transient angle stability under symmetric and asymmetric short circuit fault," *IEEE Trans. Energy Convers.*, vol. 39, no. 4, pp. 2184–2201, Dec. 2024.

[25] A. Azizi and A. Hooshyar, "Fault current limiting and grid code compliance for grid-forming inverters—Part II: Solution," *IEEE Trans. Sustain. Energy*, vol. 15, no. 4, pp. 2500–2517, Oct. 2024.

[26] R. Luo, H. Gao, F. Peng, and Y. Guo, "An adaptive low-voltage ride-through method for virtual synchronous generators during asymmetrical grid faults," *IEEE Trans. Sustain. Energy*, vol. 15, no. 3, pp. 1589–1600, Jul. 2024.

[27] N. Baeckeland, D. Venkatramanan, M. Kleemann, and S. Dhople, "Stationary-frame grid-forming inverter control architectures for unbalanced fault-current limiting," *IEEE Trans. Energy Convers.*, vol. 37, no. 4, pp. 2813–2825, Dec. 2022.

[28] H. Wu, X. Wang, and L. Zhao, "Design considerations of current-limiting control for grid-forming capability enhancement of VSCs under large grid disturbances," *IEEE Trans. Power Electron.*, vol. 39, no. 10, pp. 12 081–12 085, Jan. 2024.

[29] T. Qoria, X. Wang, and R. Kadri, "Grid-forming control VSC-based including current limitation and re-synchronization functions to deal with symmetrical and asymmetrical faults," *Electric Power Syst. Res.*, vol. 223, p. 109647, Oct. 2023.

[30] Q. Liu, K. Jia, B. Yang, L. Zheng, and T. Bi, "Fault analysis of inverter-interfaced RESs considering decoupled sequence control," *IEEE Trans. Ind. Electron.*, vol. 70, no. 5, pp. 4820–4830, May 2023.

[31] H. Wu and X. Wang, "Small-signal modeling and controller parameters tuning of grid-forming VSCs with adaptive virtual impedance-based current limitation," *IEEE Trans. Power Electron.*, vol. 37, no. 6, pp. 7185–7199, Jun. 2022.

[32] B. Fan and X. Wang, "Fault recovery analysis of grid-forming inverters with priority-based current limiters," *IEEE Trans. Power Syst.*, vol. 38, no. 6, pp. 5102–5112, Nov. 2022.

[33] A. Pal, D. Pal, and B. K. Panigrahi, "A current saturation strategy for enhancing the low voltage ride-through capability of grid-forming inverters," *IEEE Trans. Circuits Syst. II, Exp. Briefs*, vol. 70, no. 3, pp. 1009–1013, Mar. 2022.

[34] B. Fan and X. Wang, "Equivalent circuit model of grid-forming converters with circular current limiter for transient stability analysis," *IEEE Trans. Power Syst.*, vol. 37, no. 4, pp. 3141–3144, Jul. 2022.

[35] "Utilizing DER for distribution resiliency – grid-forming inverter requirements, test plan, integration with a commercial product and laboratory validation," Electric Power Research Institute (EPRI), Tech. Rep., Jan. 2023.

[36] O. Mo, S. D'Arco, and J. A. Suul, "Evaluation of virtual synchronous machines with dynamic or quasi-stationary machine models," *IEEE Trans. Ind. Electron.*, vol. 64, no. 7, pp. 5952–5962, Jul. 2017.

[37] X. He, M. A. Desai, L. Huang, and F. Dörfler, "Cross-forming control and fault current limiting for grid-forming inverters," *IEEE Trans. Power Electron.*, vol. 40, no. 3, pp. 3980–4007, Mar. 2025.

[38] L. Huang, H. Xin, Z. Wang, L. Zhang, K. Wu, and J. Hu, "Transient stability analysis and control design of droop-controlled voltage source converters considering current limitation," *IEEE Trans. Smart Grid*, vol. 10, no. 1, pp. 578–591, Jan. 2019.

[39] A. D. Paquette and D. M. Divan, "Virtual impedance current limiting for inverters in microgrids with synchronous generators," *IEEE Trans. Ind. Appl.*, vol. 51, no. 2, pp. 1630–1638, Mar. 2015.

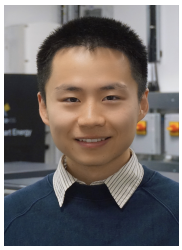
[40] "IEEE Standard for interconnection and interoperability of distributed energy resources with associated electric power systems interfaces," *IEEE Std. 1547-2018*, Apr. 2018.

[41] H. Xin, L. Huang, L. Zhang, Z. Wang, and J. Hu, "Synchronous instability mechanism of P-f droop-controlled voltage source converter caused by current saturation," *IEEE Trans. Power Syst.*, vol. 31, no. 6, pp. 5206–5207, Nov. 2016.

[42] W. Du, "Model specification of droop-controlled, grid-forming inverters (REGFM_A1)," Pacific Northwest National Laboratory (PNNL), Richland, WA (United States), No. PNNL-35110, Tech. Rep., Sep. 2023.

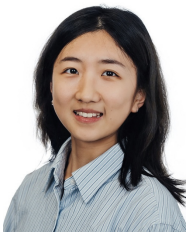
[43] X. Lyu, W. Du, S. M. Mohiuddin, and Y. Cheng, "Critical clearing time for droop-controlled grid-forming inverters with circular current limiting and virtual impedance current limiting," *IEEE Trans. Power Syst.*, vol. 40, no. 2, pp. 1997–2000, Mar. 2025.

[44] S. Chen, Y. Sun, H. Han, S. Fu, S. Luo, and G. Shi, "A modified VSG control scheme with virtual resistance to enhance both small-signal stability and transient synchronization stability," *IEEE Trans. Power Electron.*, vol. 38, no. 5, pp. 6005–6014, May 2023.



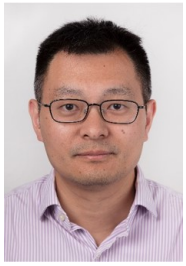
Qian Liu received the B.S. and M.S. degrees in electrical engineering from North China Electric Power University, Beijing, China, in 2020 and 2023, respectively. He is currently working toward the Ph.D. degree in electrical engineering with the School of Engineering, University of Warwick, Coventry, U.K.

He is currently a Marie Sklodowska-Curie Researcher. His research interests include modelling, stability analysis, and control design of power electronics-dominated power systems.



Miao Wang received the B.S. degree in electrical engineering from North China Electric Power University, Beijing, China, in 2020, and the M.S. degree in electrical engineering from Shanghai Jiao Tong University, Shanghai, China, in 2023. She is currently working toward the Ph.D. degree in electrical engineering with the School of Engineering, University of Warwick, Coventry, U.K. Her research interests include wind farm control, wind turbine control, and reinforcement learning.

Mostafa Nick received the Ph.D. degree in Electrical Engineering from the Swiss Federal Institute of Technology (EPFL), Lausanne, Switzerland, in 2016. From 2017 to 2024, Mostafa was with National Grid, UK, and he is currently with the National Energy System Operator (NESO), UK. His research interests include power systems operation, planning, modelling, and compliance, as well as optimisation and artificial intelligence applications in the energy sector.



Cheng Chen received the M.S. degree in power system and economics from University of Manchester in 2008. He is also a Charter Engineer (CEng) registered with the Institution of Engineering and Technology, UK. He is currently working as Head of Offline Modelling in National Energy System Operator. He has a strong track record in leading modelling development in both operational and planning environments, his main research interest is the innovative power system analysis required by Zero Carbon Operation.

Xiaowei Zhao received the Ph.D. degree in control theory from Imperial College London, London, U.K., in 2010. He was a Postdoctoral Researcher with the University of Oxford, U.K., before joining the University of Warwick, U.K., in 2013, where he is currently a Professor of Control Engineering. His research interests include control theory and machine learning, with applications in offshore renewable energy systems, smart grids, and autonomous systems.